Do Firms Innovate When the Rivals They Watch Consolidate?

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Abstract

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Competition can either strengthen or weaken firms' incentives to innovate: market power can increase the rents that reward successful research, while competitive pressure can increase the incentive to innovate in order to preserve or improve a firm's position (Schumpeter 1942; Arrow 1962; Aghion et al. 2001). Which force dominates depends on firms' competitive positions and on the strategic environment in which they invest (Aghion et al. 2005; Vives 2008; Shapiro 2012). Mergers make this tension especially consequential. By combining firms and their resources, a merger changes the competitive environment in which innovation choices are made, and can therefore reshape investment incentives both inside and outside the transaction (Federico, Langus and Valletti 2018; Motta and Tarantino 2021; Bourreau, Jullien and Lefouili 2026). These dynamic effects have become increasingly important in merger review. Competition authorities now assess not only the immediate effects of a transaction on prices and output, but also its implications for innovation, investment, potential competition, and future rivalry (Federico, Scott Morton and Shapiro 2020; U.S. Department of Justice and Federal Trade Commission 2023; Draghi 2024; Pietola, Tarantino and Zenger 2026).[1] Assessing these effects is difficult because the innovation consequences of a merger depend not only on what the merging firms will do, but also on how the firms that continue to compete with them will respond.

The empirical literature has made substantial progress on the first part of this problem. Mergers can reorganize research within the combined firm, alter the productivity of innovative activity, and, in the case of so-called killer acquisitions, lead overlapping innovation projects or pipelines to be discontinued (Bena and Li 2014; Seru 2014; Cunningham, Ederer and Ma 2021).[2] But these effects do not determine the market-wide innovation response. A merger also changes the incentives faced by firms left outside the transaction. Those firms may reduce innovative effort if competitive pressure weakens, or increase it in response to a stronger competitor. Their response can therefore amplify or offset changes in innovation inside the merged firm. Existing evidence confirms that non-merging firms do respond to consolidation, but the estimated innovation response varies across settings (Haucap, Rasch and Stiebale 2019; Kaufmann and Schiereck 2023; Li 2024; Aggarwal and Baxamusa 2025; Hu and Ma 2026). Systematic evidence on this outsider margin remains considerably thinner than evidence on the firms that merge.

A central empirical challenge is identifying which firms are meaningfully exposed to a competitor's consolidation. One approach is to assign exposure through industry membership. For example, Li (2024) defines non-merging rivals as firms sharing the merging firms' six-digit NAICS industry and shows robustness to alternative product-

[1] See, e.g., UK CMA, *Booking Holdings/eTraveli*, Phase 1 decision, 29 September 2022; European Commission, *Booking Holdings/eTraveli Group*, Case M.10615, 25 September 2023. The same transaction reached different outcomes across merger-control jurisdictions; the Commission's assessment explicitly considered the merger's effect on rivals' incentives to innovate.

[2] See, e.g., *Dow/DuPont* (Case M.7932) and *Bayer/Monsanto* (Case M.8084) on overlapping R&D capabilities and innovation competition, and *Adobe/Figma* (Case M.11033) on potential competition and reverse-killer-acquisition concerns.

market definitions. Such classifications provide a transparent ex-ante definition of exposure, but necessarily treat firms within the same narrowly defined industry as belonging to a common competitive set, even when the relevance of a particular competitor may differ across firms. More refined approaches recover rivals from merger-control investigations or from firms' position in product space. Haucap, Rasch and Stiebale (2019) and Stiebale and Szücs (2022) recover competitors from the product and geographic markets assessed by the European Commission, yielding unusually precise rival sets tied to the transaction under review. Because these relationships are recovered from the merger case itself, however, they identify the firms exposed to a particular transaction rather than a standing competitive network observed independently of merger activity; their difference-in-differences designs therefore construct comparison firms outside the case-defined rival set using propensity-score matching. Their coverage is also necessarily limited to transactions within the Commission's jurisdiction for which sufficiently detailed competitive assessments are available. A complementary approach uses firms' product descriptions to construct time-varying measures of product-market similarity (Hoberg and Phillips 2016), allowing competitive proximity to vary continuously across firm pairs and over time and increasingly being used to identify non-merging rivals in M&A settings (Deng, Javaid and Chizema 2026; Hu and Ma 2026). These measures provide a scalable and granular mapping of product-market rivalry, but the underlying similarity relation is symmetric and is constructed from the product descriptions of U.S. public firms. Across these approaches, what remains difficult to observe at scale is a competitive network that is independent of any particular transaction, directed from the perspective of each individual firm, and able to include competitors globally.

We address this problem by observing competitive relationships from firms' own disclosures over time. FactSet Revere records the competitors that a firm itself identifies, allowing competitive exposure to be defined at the firm–rival level rather than assigned from industry membership. The resulting rival sets are selective and asymmetric: the median firm names 8 rivals, only 13.0 per cent of named relationships are mutual, and 43 per cent connect firms in different two-digit SIC industries. We combine these links for 17,361 firms in 102 countries over 2003 to 2022 with control-changing transactions from SDC Platinum, matching on the self-designated rival. Because the links are directed, we can distinguish the consolidation of a competitor the focal firm itself names from the consolidation of a firm that names the focal firm but is not named in return. This setting allows us to ask not only whether non-merging firms respond to rival consolidation, but whether the response depends on *whose* consolidation the firm is exposed to.

We find that firms increase research spending after a named rival consolidates. Among firms reporting positive R&D, research spending is about 4 per cent higher in the year of the onset, 10 per cent higher one year later, and 24 per cent higher three years later, relative to the year before the onset and to the contemporaneous path of comparison firms.

The response becomes substantially larger by the fifth year.[3] This increase occurs alongside an expansion of the firm: assets and employment also rise. Research intensity responds more slowly, with R&D relative to lagged assets remaining approximately unchanged through the fourth year and increasing only at the fifth horizon. The early response is therefore primarily an increase in research spending as firms expand.

The increase in research spending is not accompanied by higher patent volume within the first five years. Firms remain approximately as likely to originate a patent family, but originate fewer families in the second and third years after a rival's consolidation. At the same time, individual patent families span more technology subclasses, while the number of distinct subclasses in which the firm patents remains essentially unchanged. Patenting therefore becomes concentrated in fewer, broader inventions rather than expanding into new technological areas. This pattern is consistent with additional research taking longer to generate patentable output, with changes in how firms organize or bundle their inventions, or with additional spending not translating into greater measured output. Estimates at longer horizons provide suggestive evidence for the first interpretation.[4]

The more distinctive result is that the response depends on which competitive relationship links the firm to the consolidating rival. R&D rises when the consolidating firm is a rival the focal firm itself names, including when the relationship is mutual, but not when the naming relationship runs only in the opposite direction. This directional difference remains among very large consolidating rivals, so it is not simply a consequence of rival size. Nor does the response systematically increase with the share of the rival set that consolidates, with horizontal deals, with proximity in industry or product space, or during merger waves. Most starkly, applying the same design to four-digit industry peers produces no comparable increase in R&D in the first three years after the deal. The definition of competitive exposure therefore materially changes the estimated innovation response to consolidation.

Finally, the response is not confined to a narrow group of firms. It appears earlier and more clearly among smaller firms, firms with less accumulated research capital, and firms with lower leverage. Taken together, these patterns point to earlier adjustment among firms with fewer internal resources when a rival they identify consolidates.

Our paper contributes to the literature on the effects of mergers beyond the firms directly involved in the transaction. Early work used the stock-market reactions of non-merging rivals to infer the competitive consequences of mergers (Stillman 1983; Eckbo 1983; Song and Walkling 2000; Fee and Thomas 2004; Shahrur 2005; Clougherty and Duso 2009; Akdoğu 2009). More recent work has moved from valuation effects to firms' real responses, including markups, investment, employment and, increasingly, innovation

[3] The fifth-year estimate is $+0.95^{***}$ log points and is based on fewer, earlier events. The same rising pattern appears when we hold the horizon sample fixed.

[4] Beyond the main five-year horizon, estimates for originated patent families turn positive in years seven and eight; granted patents also rise, and distinct technology subclasses increase by year eight.

(Filson, Olfati and Radoniqi 2015; Stiebale and Szücs 2022; Li 2024). Within this emerging merger–innovation literature, studies document changes in rivals' R&D and patenting (Haucap, Rasch and Stiebale 2019; Ahmad, de Bodt and Harford 2020; Kaufmann and Schiereck 2023; Xue et al. 2024; Aggarwal and Baxamusa 2025), as well as in the direction, breadth and economic value of their inventive activity (Valentini 2016; Hu and Ma 2026; Deng, Javaid and Chizema 2026). We add to this literature by studying the dynamic research response to rival consolidation across industries and countries, and by making the mapping between a merger and the outsiders exposed to it part of the empirical question. The directed competitor relationships allow us to distinguish which firms regard the consolidating company as a rival, rather than treating competitive exposure as common to all firms occupying the same market or product space.

This contribution is directly relevant to merger policy. The innovation consequences of a transaction depend not only on what happens inside the merged firm, but also on how the firms left outside it adjust their own research and inventive activity. Recent evidence and policy work increasingly recognize this outsider margin (Haucap, Rasch and Stiebale 2019; Stiebale and Szücs 2022; Aggarwal and Baxamusa 2025; Hu and Ma 2026). Our evidence adds that identifying the relevant outsiders is itself consequential: competitive responses can be firm-specific and asymmetric, and need not be confined to firms inside a conventional industry boundary. This raises further questions about when outsider responses offset or reinforce innovation changes within the merged firm, which competitive relationships transmit those effects, and how such firm-level adjustments aggregate into the market-wide innovation effects ultimately relevant for merger review.

The rest of the paper is organized as follows. Section I presents the empirical design and Section II the data. Sections III to V report the response of research spending, whose consolidation matters and which firms respond; Section VI examines alternative explanations, and Section VII concludes.

I. Empirical design

A. The exposure mapping

We study how a firm's research spending responds when one of its rivals enters a consolidation, a deal that transfers control of a company or of its assets. The deal is the rival's and the outcome is the firm's: the firm is not itself treated. Let M_{jt} indicate that company j enters a consolidation in year t, and let $\mathbf{M}$ collect these indicators over all companies and years, in our sample or not. The research spending of firm i in year $t+h$ can respond to any of these deals, so its potential outcomes are $Y_{i,t+h}(\mathbf{M})$ (Manski 2013), and the usual assumption that a unit's outcome depends only on its own treatment does not apply.

An exposure mapping reduces $\mathbf{M}$ to what matters for each firm (Aronow and Samii

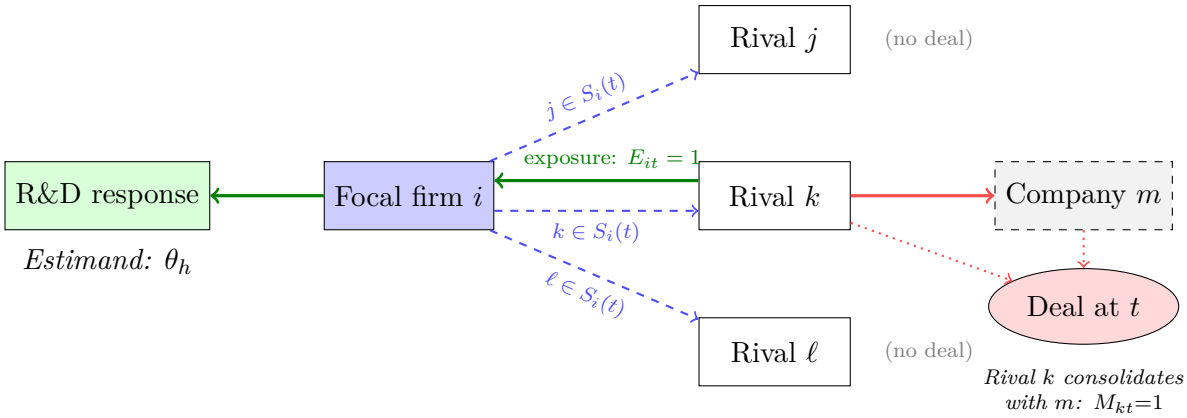


Figure 1. The main exposure mapping

Notes: Focal firm i names rivals j, k and ℓ as competitors in year t; they form its reference set $S_i(t)$ (dashed arrows, in the direction of naming). Rival k enters a consolidation with company m in year t, as the buyer or as the company bought ($M_{kt} = 1$); m can be any company. The deal exposes firm i ($E_{it} = 1$, equation (1)). If no rival of i consolidated in $t-1$, year t is an onset, and θ_h (equation (2)) is the average change in the R&D of the firms at an onset from $t-1$ to $t+h$, less the change in the quiet state.

2017). Ours has three parts. The first is a reference set $S_i(t)$, the companies whose deals reach firm i in year t. In the main design, it holds the firm's *designated rivals*, the companies its own disclosures name as competitors in year t. The second is the deals that count, here every consolidation by a member of the set. Together they define the firm's exposure:

$$E_{it} \;=\; \mathbf{1}\Big[\textstyle\sum_{j\in S_i(t)} M_{jt} \;\geq\; 1\Big]. \tag{1}$$

The third part is the rule that turns exposures into events and comparisons, set out in Sections I.B and I.C. The firm's own deals, M_{it}, are left out of the mapping and enter the controls. Figure 1 draws the main mapping.

The mapping defines the effect we estimate. For horizon h, it is the average change in research spending of the firms at an onset, from the year before the onset to year $t+h$, less the change the same firms would have seen in the quiet state:

$$\theta_h \;=\; \mathrm{E}\Big[\,(Y_{i,t+h} - Y_{i,t-1}) - (Y^0_{i,t+h} - Y^0_{i,t-1}) \;\Big|\; D_{it} = 1\,\Big], \tag{2}$$

where D_{it} indicates an onset and Y^0_{is} is the firm's research spending in the quiet state, in which no member of its reference set consolidates from $t-5$ to $t+h$. Because an episode continues after its first deal, θ_h includes the effect of the deals that follow the onset. For a firm whose rivals consolidated a few years before $t-1$, it also includes any response to those earlier deals that is still under way.

Following Sävje (2024), we treat the mapping as a definition, separate from the assumptions that give it a causal reading. The mapping does not need to describe how deals actually reach firms. A firm may respond to the deals of companies it does not

name, or differently to different deals of those it does; θ_h is then still well defined, as an average over the deals the mapping leaves out and over those it groups together. Because the mapping is a definition, the rest of the paper varies it one part at a time while the question stays the same (Appendix Table B1). Changing the reference set asks whose consolidation matters; splitting the exposed state by the features of the deals asks which consolidations matter; changing the deals that count, their dating or the length of the quiet state tests the definition itself; and adding the exposures of other mappings to the controls asks whether the response to the rivals' deals survives them.

B. The event

Exposure is repeated, not absorbing: a firm's rivals consolidate many times over two decades, often in consecutive years. The event in year t is therefore the *onset* of a consolidation episode, $D_{it} = E_{it}\,(1 - E_{i,t-1})$: at least one designated rival consolidates in t, and none did in $t-1$. The firm is followed through the deals that come after the onset, and a new event begins only after another year without a rival deal.

The estimator follows an exposed firm only until its event indicator turns on again. If every year with a deal were an event, each follow-on deal would end the episode, and the later horizons would keep only the episodes in which the rivals stopped dealing, a selected sample. With the onset as the event, follow-on deals belong to the episode, and we estimate the response to an episode of rival consolidation from its first deal on.[5]

C. The comparison group

For horizon h, a comparison firm-year is one in the quiet state: no designated rival consolidates in year t, in any of the five years before it, or through $t+h$. The forward condition is the estimator's own. The backward condition is ours. The estimator's standard comparison requires the rivals to be quiet only in $t-1$ and t, and so admits firms whose rivals consolidated two or three years earlier. If the response takes years to build, such firms may still be raising their research spending during the window we measure, which would bias the estimate toward zero without showing in the placebo years. The five-year condition applies to the comparison firm-years only; imposing it on the events would keep only the onsets that follow five quiet years.

D. Estimation

We estimate the response with the local-projection difference-in-differences estimator of Dube et al. (2025), in the Stata implementation of Busch and Girardi (2023). For each

[5] Taking the firm's first exposure as an absorbing event would use one episode per firm, and the first exposure is rarely observed, since most firms already have a consolidating rival in their first year in the sample. Treating exposure as a status that switches on and off, the estimator's default, lets continuously exposed firms serve as comparisons on the premise that their response settles within a year.

horizon h from -4 to 5, the regression is

$$(3) \quad y_{i,t+h} - y_{i,t-1} \;=\; \beta_h\, D_{it} \;+\; \rho_h\, \Delta y_{i,t-1} \;+\; \mathbf{x}'_{i,t-1}\boldsymbol{\gamma}_h \;+\; \alpha^h_{s(i),t} \;+\; \alpha^h_{c(i),t} \;+\; \varepsilon_{i,t+h},$$

where y_{it} is the outcome, $\Delta y_{i,t-1}$ its change in the year before the onset, $\mathbf{x}_{i,t-1}$ the firm's characteristics, and $\alpha^h_{s(i),t}$ and $\alpha^h_{c(i),t}$ two-digit-industry-by-year and country-by-year fixed effects. The regression for horizon h keeps only the events and comparison firm-years that are clean through $t+h$, so that no firm already exposed serves as a comparison, as it would in a two-way fixed-effects regression with staggered exposure and effects that change over time.[6] The events are reweighted to carry equal weight at every horizon, so that β_h is the average change in the outcome from $t-1$ to $t+h$ relative to comparison firms of the same industry, country and year; under the assumptions of Section I.F, it estimates θ_h. As a long difference, it removes any difference in levels between exposed and comparison firms.

The year before the onset is the reference, and $h = -4$ to -2 are placebo years. Because the lagged change in the outcome is a control, the estimate at $h = -2$ is zero by construction, and the informative placebo years are $h = -4$ and $h = -3$.[7] We also report the pooled post-period estimate, the average from $h = 0$ to $h = 5$ on the firm-years observed through $h = 5$, and the pooled estimate over the placebo years. A firm-year enters the regression for horizon h only if its outcome is observed at $t-1$ and $t+h$ and its controls at $t-1$ and, through the growth terms, at $t-2$; the later horizons therefore rest on fewer and earlier events.

E. Controls, fixed effects and inference

The controls, all measured before the onset, form four blocks. The *firm* block holds log total assets, log sales, log age, and cash and capital expenditure over assets at $t-1$, which bear on how fast research spending grows. The *growth* block holds the growth of assets and sales from $t-2$ to $t-1$, which, with the lagged change in the outcome, fixes the path the firm was on before the onset. The *rival-set* block holds the number of live designated rivals at $t-1$ and indicators for a year without a rival on record at t and $t-1$. The chance that some rival consolidates rises with the number of rivals, so a firm with many rivals is rarely quiet for five years, and the comparison group leans toward firms with few rivals or none on record; this block compares firms whose rival sets have the same size and are recorded alike. The *own-deal* block holds indicators for the firm's own consolidations at t, $t-1$ and $t-2$. A firm that acquires or is acquired changes its research spending for reasons of its own, including the consolidation of a target's spending into its accounts. These deals are controlled for rather than excluded, so that the events are not selected on

[6] See de Chaisemartin and D'Haultfœuille (2020), Goodman-Bacon (2021), Callaway and Sant'Anna (2021) and Sun and Abraham (2021), and Roth et al. (2023) for a review.

[7] At $h = -2$ the outcome is $y_{i,t-2} - y_{i,t-1} = -\Delta y_{i,t-1}$, which the control absorbs exactly.

the firm's own behaviour, and its deals after t are left in, since they may be part of its response.

The fixed effects remove the shocks common to an industry and year and to a country and year, including national differences in how research spending is reported and in when data coverage begins. Standard errors are clustered by the firm's reporting unit, the firm itself or the group of entities that report one set of accounts.[8] In the text, *, ** and *** denote significance at the 10, 5 and 1 per cent levels; the tables and figures report the standard errors.

F. What the design assumes

The estimator recovers θ_h under five assumptions.

Assumption 1 (exposure). Given its own deals, a firm's research spending depends on consolidations only through its exposures E_{is}. In its strict form, the assumption requires *exclusion*, that the deals of companies outside the reference set do not move the firm's research spending, and *sufficiency*, that which members of the set consolidate, how many and in what deals matter only through whether any does. The design does not need the strict form. Without sufficiency, θ_h averages the effects of the deals the mapping groups together, and Section IV splits them. Exclusion is needed only in a weak form (Leung 2022; Sävje 2024): the deals the mapping leaves out may move a firm's research spending, but not differently in exposed and comparison firm-years with the same controls. The weak form fails in two ways. Comparison firms may be reached by the same deals through links the mapping omits, when the consolidating company names them or competes with them unnamed, which biases the estimate toward zero (Butts 2023). And the deals of the firm's suppliers, customers, partners or industry may coincide with its rivals' deals, so that the estimate attributes their effect to the rivals; such spillovers are common in corporate finance, and fixed effects can worsen the bias they cause (Berg, Reisinger and Streitz 2021). Section VI.A controls for each of these deals and takes each as an event.

Assumption 2 (parallel trends in the quiet state). Had their rivals stayed quiet from $t-5$ through $t+h$, the exposed firms' research spending would have changed from $t-1$ to $t+h$ as that of comparison firms with the same characteristics, in the same industry, country and year. Under interference, comparison units may themselves be exposed to the treatment of others, so the assumption must be restated (Xu 2025, 2026); we state it for the quiet state, which the comparison firms occupy by construction. The timing of the event follows the rival's decision to buy or to be sold, which reflects its own strategy, its financing and the market for deals; the firm's research spending responds to its own

[8] Interference makes outcomes dependent along the network through which it runs (Leung 2022): firms that name the same rival are exposed to the same deal. A cluster built on that link is not available, since a firm names several rivals that are in turn named by different firms. Section VI.F reports other clusterings and randomization inference.

opportunities and finances and to shocks to its industry and country. The assumption fails if the exposed firms were already on a different path, which the placebo years, the balance of the two groups and the paths of their characteristics test (Section III.B).

Assumption 3 (no anticipation). Research spending does not respond to the rivals' deals before the onset. Deals are dated at their announcement, their first public date; the assumption fails if firms learn of a deal earlier and invest ahead of it. The placebo years would show such a response, and Section VI.B takes rumoured and withdrawn deals as events.

Assumption 4 (limited carryover). A firm whose rivals have made no deal for five years no longer responds to their earlier deals. Section VI.G lengthens the quiet window to seven years and restricts the comparison to firms not yet exposed.

Assumption 5 (predetermined reference sets). The reference set does not respond to the deals it measures: a firm does not name a competitor because it expects the competitor to consolidate, nor because of its own research plans. Studies of interference take the network as given (Aronow and Samii 2017; Forastiere, Airoldi and Mealli 2021; Leung 2022); ours comes from the firms' own disclosures and could respond to both. Section VI.D follows the rival set around the onset and compares old and new ties.

II. Data

A. Competitor relationships

The competitors a firm names come from FactSet Revere Supply Chain Relationships (FactSet Research Systems 2021*b*). FactSet's analysts read a listed company's public documents, including its annual reports, regulatory filings, investor presentations, press releases and website, and record the companies these documents identify as competitors, customers, suppliers or strategic partners. Each record has a type, a *source* company, whose documents were read, a *target* company, which they identify, and the dates over which FactSet held it current. The data run from 3 April 2003 to 23 November 2021; their 414,858 competitor records were written by 18,216 source firms in 104 countries and name about 75,902 companies. A firm's designated rivals are the targets of its competitor records, and the *focal firms*, whose research spending we study, are those that name at least one competitor.[9]

The dates of a record are those of the database entry, not of the relationship: FactSet re-publishes many records under new entries and closes others in batches during database maintenance. We therefore read relationships by calendar year. A firm and a rival are related in every calendar year touched by one of their records, and a firm's rival set in a

[9] The other types of record define the channels of Section VI.A: the deals of a firm's suppliers and customers, of its partners, and of the rivals of its rivals.

year holds the companies named in its records live in that year.[10] A firm's coverage starts with its first record as a source, of any type; its earlier years do not enter the sample. Within its coverage, a year in which none of its competitor records is live is either a year in which its documents named no competitor or one in which FactSet did not read them. Such years make up 7.3 per cent of the estimation firm-years. We code them as years without a rival deal and control for them with indicators at t and $t-1$. A recorded deal is never in doubt; the absence of one can be. In a comparison firm-year, a missed rival deal makes the comparison firm respond like an exposed one and biases the estimate toward zero; at an event, it means that the onset may fall inside an episode. Treating these years as missing is not feasible: the estimator keeps a comparison firm-year at horizon h only if its status is observed in every year from $t-1$ to $t+h$, and too few would remain at the later horizons.

The competitor records have the features of attention. The sets are small and stable: in the years in which a firm names competitors, its set holds a median of 8 companies (a mean of 10.2 and a ninetieth percentile of 22), a firm keeps on average 90 per cent of the rivals it named the year before, and a named pair lasts a median of 3 years. Naming is rarely mutual: only 13.0 per cent of the directed pairs exist in both directions. Naming points upward (Figure 2): 76 per cent of the designated rivals have higher sales than the firm that names them, and 83 per cent of those that do not name it back, by a median factor of about 12; mutual rivals are close in size, with a median gap of 1.0 log points in sales against 2.7 for pairs named one way. And naming crosses the boundaries along which markets are usually drawn: 43 per cent of the named pairs join firms in different two-digit SIC industries, and 46 per cent firms in different countries.

Research on competitive dynamics attributes selective and asymmetric naming of this kind to attention: a firm can watch a competitor that does not watch it back (Chen 1996; Chen, Su and Tsai 2007). We read a firm's designated rivals as the competitors it watches. Firms also choose which competitors to name in order to present themselves (Eisdorfer, Ozik and Sadka 2023), but the records draw on all of a firm's public documents rather than on one filing, and their persistence from year to year makes them a poor instrument of positioning.

The text-based network industries of Hoberg and Phillips (2016), from their data library (Hoberg and Phillips 2026), provide an independent benchmark. They relate two US-listed firms in a year when the product descriptions of their 10-K filings are sufficiently similar, so the network reflects what firms say about their own products rather than about their rivals. Where both firms of a named pair are in the network's universe, as for 22 per cent of the pairs, 71 per cent of the named pairs are related in its embeddings-based

[10] The median competitor record lasts 1.6 years, and 21 per cent of the named pairs appear in several records, mostly re-publications of the same relationship. The Supplementary Appendix describes two other readings, used as robustness checks: one bridges the gaps between a pair's records, and one counts a deal only within the exact dates of a record.

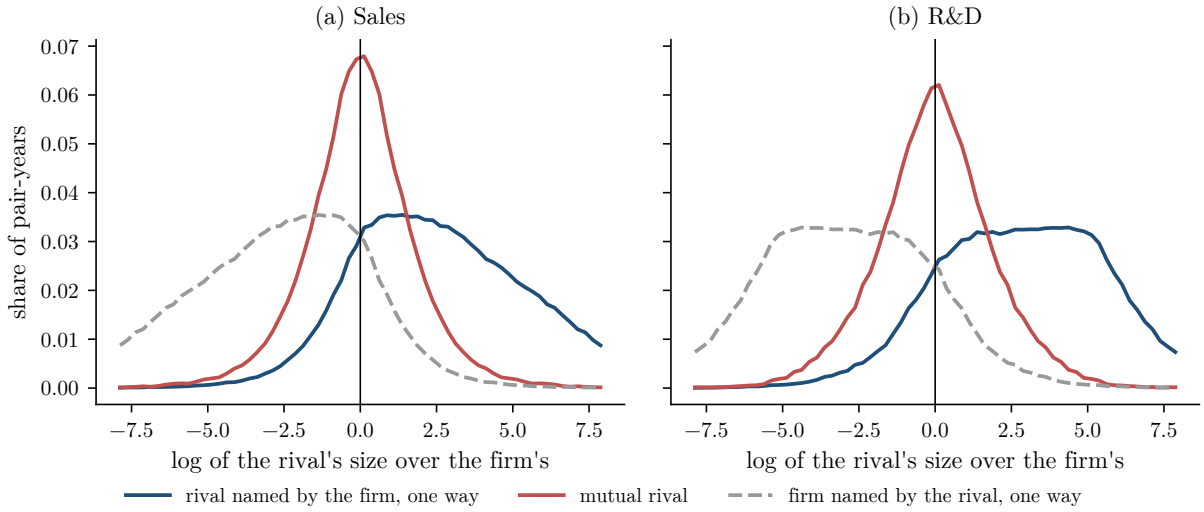


Figure 2. Whom firms name

Notes: The size of a named rival relative to the firm, by the direction of naming. Distribution of the log ratio of the rival's size to the firm's, in sales (panel a) and in R&D (panel b), over the designated rival pair-years with accounts for both firms, 2003–2021. A pair named in one direction appears twice: once from the side of the firm that names (solid line, "rival named by the firm") and once from the side of the firm that is named (dashed line, "firm named by the rival"). A mutual pair, in which each firm names the other, is symmetric by construction, so that only its spread is informative.

version (Appendix Table C1), against 52 per cent of the pairs of our firms that share a four-digit SIC code, 41 per cent of those that share a three-digit code and 3 per cent of random pairs. Mutual pairs are related in 91 per cent of cases and pairs named one way in 65 per cent.[11] Named rivals are a selection among the firms with similar products: a firm is related to a median of 101 firms in the network in a year and names 3 of them. They also reach beyond the network: 42 per cent of our pairs have a naming firm outside the United States, 49 per cent a rival outside it and 42 per cent a rival that is not listed, and the network observes only 33 per cent of the firm-rival pairs in our events.

B. Consolidations

The deals come from Refinitiv's SDC Platinum mergers and acquisitions database (LSEG Data & Analytics 2026), accessed through WRDS: 1,348,776 deals announced from 1999 to 2026, 925,859 of them from 2003 to 2021. SDC identifies a deal's parties by name, nation and six-digit CUSIP, a securities identifier that is reassigned over time and that SDC estimates for private companies, whereas FactSet identifies entities. We attribute each deal to the FactSet entity that took part in it, combining matches on dated security identifiers with matches on names, which yields 484,628 links between an entity, a deal

[11] With the word-based version used in most published work, the agreement is lower and ordered in the same way: 60 per cent of the observable named pairs are related, against 51 per cent of the pairs that share a four-digit code, 41 per cent of those that share a three-digit code and 3 per cent of random pairs.

and the entity's role in it.[12] A deal belongs to a firm only when the firm itself is a party: a subsidiary's deal is not its parent's, and the sale of a division is a divestiture by its owner.

A *consolidation* is a completed deal that transfers control of a company, a division or a set of assets, with the rival as the buyer or as the company bought, dated at its announcement. Buybacks and deals within a corporate group are excluded. Withdrawn deals, rumours, pending deals, minority stakes and partial purchases of assets are recorded separately and never count as consolidations. A rival's consolidation reaches a firm in the fiscal year that contains the announcement date if the rival is in the firm's set in that calendar year. From 2003 to 2021, the designated rivals of the focal firms entered 101,308 consolidations, 87,847 as the buyer and 13,461 as the company bought, and 34 per cent of the designated rivals consolidated at least once. A value is disclosed for 50 per cent of these deals, with a median of 78 million dollars; 35 per cent cross a border, and in 13 per cent the company bought is listed.

The attribution passes an independent check on the company bought. Of the focal firms that Compustat stopped following because they were acquired, 85 per cent are the company bought in one of our consolidations within two years of that date.

C. Accounts and research spending

The financial accounts come from S&P's Compustat North America and Global (S&P Global Market Intelligence 2026), completed by FactSet Fundamentals (FactSet Research Systems 2021*a*) and Bureau van Dijk's Orbis (Bureau van Dijk 2026) for the years Compustat leaves empty (2,384 and 825 firm-years). No table links FactSet's entities to Compustat's companies, and the two vendors follow opposite conventions at a merger: Compustat keeps one company's record for the combined accounts, while FactSet opens a new entity and retires the old one. We built a crosswalk that holds year by year, combining matches on dated security identifiers and on names; it reaches 96.9 per cent of the focal firms. Each firm-year takes all its variables from one vendor, so that no ratio mixes two vendors' definitions. Where two vendors report the same firm and fiscal year, their total assets agree within five per cent in 96.3 per cent of cases, their sales in 94.2 per cent and their R&D in 82.2 per cent. Amounts are in US dollars, converted at the period-end rate for balance-sheet items and the period-average rate for flows. Of the firm-years, 58 per cent report under US GAAP, 32 per cent under IFRS and 10 per cent under a local standard.

Three choices define the outcome. First, R&D is net of acquired in-process R&D. Until 2008, a US acquirer expensed at once the part of a purchase price attributed to the target's unfinished research, and Compustat's R&D item includes this charge, which

[12] The Supplementary Appendix (`https://salma-lemkhente.github.io/innovation-rival-consolidation/Appendix/Supplementary_Appendix.pdf`) describes how the deals, the accounts and the patents are matched to the firms.

would raise an acquirer's R&D mechanically in the year it buys; we subtract it in the 2,539 firm-years that carry it. Second, a missing value is never treated as a zero, since a firm that reports no R&D may still do research.[13] R&D is positive in 48.9 per cent of the estimation firm-years, zero in 6.5 per cent and missing in 44.6 per cent. Third, the outcome takes three forms: the logarithm of R&D in the firm-years with positive R&D, our main outcome; its inverse hyperbolic sine, which keeps the reported zeros;[14] and R&D over the total assets of the year before, which measures research effort relative to the firm's size.

D. Patents

The patents come from Google's table of patent publications (IFI CLAIMS Patent Services and Google 2026), which covers every patent office. Patents carry the name of their assignee but no firm identifier, so we validate and date assignee names rather than individual patents: a name is attributed to a firm for the years in which it denoted the firm or a company the firm controlled. The candidate names come from published links between patents and firms, such as those of Kogan et al. (2017), and from a search among the assignees of all offices for the names these links miss, such as former names, the names under which firms file at home and the names of subsidiaries; as for the deals and the accounts, the matching combines identifiers, where the links carry them, with names. Of the focal firms, 9,643 (53 per cent) have a validated name, and they account for 91 per cent of the firm-years with positive R&D. A firm without a validated name is a firm whose patents we cannot see, not a firm without patents, so its patent outcomes are missing rather than zero; 61 per cent of the estimation firm-years have them.

We count the patent families the firm originates, by priority year, setting aside the 6.4 per cent of families that reached it through an acquisition; its granted patents, by filing year; and the distinct technology subclasses (CPC) of its families in the year, which measure the breadth of its patenting. The counts enter as inverse hyperbolic sines. Because grants come two to four years after filing, the counts of granted patents for the last filing years are incomplete; the shortfall is common to the firms filing in the same year and is absorbed by the year effects.

E. The sample

The estimation sample holds the focal firms' fiscal years between their first and last competitor record in which the firm has accounts of its own and an observed rival set:

[13] FactSet records a zero where Compustat has no value in a share of firm-years that changes with the vintage of FactSet's collection rather than with firms' research. A FactSet zero is therefore accepted only where Compustat shows that the firm never reports positive R&D or does not cover it.

[14] The inverse hyperbolic sine is close to the logarithm for large values and close to the identity near zero, so its estimates depend on the unit of measurement (Bellemare and Wichman 2020; Chen and Roth 2024). R&D is measured in millions of dollars.

Table 1. Summary statistics of the estimation sample

	Firm-years	Mean	Standard deviation	25th percentile	Median	75th percentile
R&D						
R&D (million dollars)	58,548	156.27	810.89	1.99	12.22	52.28
Log R&D	51,726	2.80	2.18	1.43	2.84	4.15
Inverse hyperbolic sine of R&D	58,548	3.17	2.19	1.44	3.20	4.65
R&D over lagged assets	58,470	0.113	0.202	0.006	0.037	0.127
The firm						
Log total assets	105,600	6.66	2.33	5.12	6.65	8.17
Log sales	99,645	5.99	2.46	4.54	6.11	7.63
Log age	103,638	3.29	0.87	2.77	3.26	3.87
Log employment	89,875	7.23	2.31	5.61	7.30	8.90
Cash over assets	102,902	0.213	0.233	0.047	0.122	0.292
Capital expenditure over assets	101,187	0.039	0.048	0.008	0.023	0.052
Leverage	105,023	0.238	0.235	0.041	0.190	0.361
Rivals and deals						
Live designated rivals	105,678	10.58	10.22	4.00	8.00	14.00
A rival in a control-changing deal	105,678	0.740	0.439	0.000	1.000	1.000
No rival on record	105,678	0.073	0.260	0.000	0.000	0.000
Own control-changing deal	105,678	0.233	0.423	0.000	0.000	0.000
Patents (firms matched to patent assignees)						
Patent families originated	64,844	46.62	348.05	0.00	1.00	8.00
Patents granted	64,844	68.63	441.72	0.00	2.00	15.00
Technology subclasses	64,844	8.60	23.65	0.00	1.00	6.00

Notes: The estimation firm-years, fiscal years 2003 to 2022: firms that name competitors in FactSet's relationship data, in the years their rival set is observed, with accounts. R&D is missing where the firm does not report it and zero where it reports none; the log is taken of positive R&D, the inverse hyperbolic sine of every reported amount. Ratios winsorised; leverage is debt over total assets. Live designated rivals: the rivals the firm names with a record live in the year; a rival in a control-changing deal: at least one of them acquires or is acquired in a deal that changes control of a firm, a division or assets; no rival on record: a year inside the firm's coverage with no live rival recorded; own control-changing deal: the firm itself acquires or is acquired. Patents: the patent families the firm originates, by priority year; its patents granted, by filing year; the distinct CPC subclasses of the year's families.

105,678 firm-years of 17,361 firms in 102 countries, over the fiscal years 2003 to 2022. R&D is reported in 58,548 of these firm-years, by 9,902 firms, and is positive in 51,726, with a median of 17.1 million dollars where positive (Table 1). The median firm-year names 8 rivals, and in 23 per cent of the firm-years the firm itself acquires or is acquired.

Rival deals are frequent and come in runs. In 74 per cent of the firm-years, at least one designated rival enters a consolidation. The sample holds 10,111 onsets, and of those observed a year later, 74.2 per cent are followed by another year of rival deals. The first exposure is rarely observed: 73.7 per cent of the firms already have a consolidating rival in their first year in the sample. The quiet state holds 13,338 firm-years without a rival deal in the year or in the five before it, and the main regression for log R&D compares 3,588 events with 3,427 comparison firm-years at $h = 0$.

Table 2 compares the sample with every company that has accounts in FactSet Fundamentals, which covers listed companies worldwide under one set of definitions, over

the fiscal years 2003 to 2021, with FactSet's figures in all three columns. The sample holds 18 per cent of these companies and 11 per cent of their company-years, but 38 per cent of their sales and 58 per cent of their R&D. Its firms are larger than the typical listed company, with median total assets of 781 million dollars against 122 million; they report R&D more often (56 against 43 per cent of company-years) and are more often manufacturers (43 against 37 per cent). The sample leans toward North America, with 60 per cent of its firm-years against 24 per cent of the universe's, while Asia and the Pacific account for 18 against 48 per cent, because FactSet covered North America from the start and the other regions later. The sample covers 82 per cent of the R&D reported by North American companies, 60 per cent of that of European companies and 25 per cent of that of the companies of Asia and the Pacific.

The sample thus represents corporate research better than it represents companies. It consists of the large, research-intensive listed firms that account for most of the R&D of listed companies, in North America and Europe above all, and the estimates describe how these firms respond. They say less about small firms and about the firms of Asia, whose coverage begins later. Because the design compares firms within country and year, the different timing of coverage across regions does not enter the comparisons.

III. Do firms raise research spending when a rival they watch consolidates?

A. The response of research spending

Figure 3 plots the estimates of equation (3). Relative to the year before the onset and to the comparison firms, the log R&D of the exposed firms is $+0.04^{*}$ higher in the year of the onset, $+0.10^{**}$ a year later, $+0.21^{**}$ by the third year and $+0.95^{***}$ in the fifth. The exposed firms thus spend about 4 per cent more on research in the year of the onset, 10 per cent more a year later and 24 per cent more by the third year. The inverse hyperbolic sine, which keeps the firms that report no R&D, gives the same profile at a smaller scale, significant only in the fifth year ($+0.37^{**}$). Almost every firm that reports no R&D before the onset still reports none years later, exposed or not, so the response comes from firms that already do research.[15]

The fifth-year estimate is the largest and the least precise. Its 95 per cent confidence interval runs from 0.42 to 1.48, and it rests on 785 observations against 7,015 at the onset, because only the earlier events can be followed for five years. The pooled average over the six years, $+0.49^{***}$, is estimated on the firm-years observed throughout and gives the fifth year the weight of the first; we do not read it as the average response. The rise does not

[15] With R&D in thousands of dollars rather than millions, the pooled response of the inverse hyperbolic sine is $+0.42^{**}$, and with R&D in dollars $+0.44^{*}$, with pooled placebo estimates of -0.02 and -0.02.

Table 2. The sample against the companies with FactSet Fundamentals accounts

	(1) Universe	(2) Name competitors	(3) Estimation sample
Coverage			
Companies	88,426	17,603	16,162
Company-years	855,410	209,108	95,236
Countries	153	102	101
Share of the universe's total (%)			
Company-years	100.0	24.4	11.1
Sales	100.0	61.4	38.3
Total assets	100.0	50.8	30.3
Employees	100.0	55.2	33.8
R&D	100.0	75.6	57.7
Company-years by region (%)			
North America	24.0	42.4	60.0
Europe	18.7	18.4	14.2
Asia and the Pacific	48.3	31.0	18.0
Rest of the world	9.0	8.2	7.9
Share of the region's R&D (%)			
North America	100.0	91.0	82.3
Europe	100.0	80.3	59.9
Asia and the Pacific	100.0	52.3	24.6
Rest of the world	100.0	77.2	68.0
Company-years by industry (%)			
Manufacturing	37.4	42.3	43.0
Services	14.9	15.1	17.1
Finance, insurance and real estate	20.4	14.4	14.8
Wholesale and retail trade	7.0	6.9	6.8
Transport, communications and utilities	7.8	10.7	10.8
Agriculture, mining and construction	12.3	10.7	7.5
The median company-year			
Total assets (USD millions)	122	489	781
Sales (USD millions)	65	272	416
Employees	558	1,166	1,391
Reports R&D (%)	43.2	52.0	56.3
R&D when positive (USD millions)	2.9	9.0	17.4

Notes: Fiscal years 2003 to 2021. (1) Every company with FactSet Fundamentals accounts and positive total assets in the year. (2) Those that name competitors in FactSet's relationship data in any year. (3) The firm-years of the estimation sample: firms that name competitors, in the years their rival set is observed, with accounts. All three columns use FactSet Fundamentals values, so that they are measured alike; 95,236 of the sample's 105,053 firm-years to 2021 are found there (the others take their accounts from Compustat or Orbis), so column (3) understates the sample's coverage slightly. Region: the company's headquarters country, grouped as in FactSet's coverage of relationships; industry: the divisions of the company's primary SIC code. Sales, assets and R&D in US dollars; totals over the years and companies of the column.

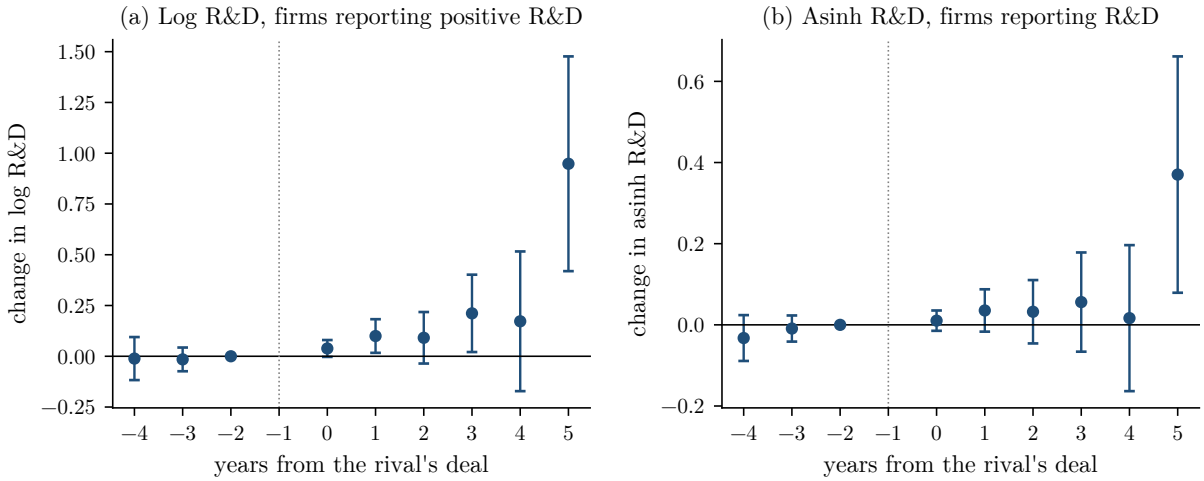


Figure 3. The response of research spending to a watched rival's consolidation

Notes: Each point is the estimated change in the firm's R&D from the year before the deal ($h = -1$) to year h, for the firm-years in which a rival the firm names enters a control-changing deal after a year without one (3,588 events in panel (a), 4,328 in panel (b)), against firm-years whose rivals make no such deal in the five years before and through the horizon. Local-projection difference-in-differences with industry-by-year and country-by-year fixed effects and the firm controls of Section I; bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit. Horizons −4 to −2 are placebo horizons; the estimate at −2 is zero by construction, because the lagged change in the outcome is a control.

come from which firms can be followed that long. Exposure does not make a firm more likely to remain observed: the probability that its R&D is reported in every year to $t + 5$ is +1.5 percentage points higher for the exposed firms, against a rate of 31 per cent among the comparison firms. And on the balanced sample of events and comparison firm-years observed at every horizon (1,083 events), the response builds in the same way, to $+0.14^{*}$ in the first year, +0.22 in the third and $+0.75^{***}$ in the fifth (Appendix Figure E1).

The data do not pin down how far the response has grown by the fifth year. The estimates of the first years barely move when the specification changes, but those of the later years do (Appendix Table E3). Without the country-by-year effects, the response is $+0.07^{*}$ in the first year, $+0.18^{**}$ in the third and $+0.45^{**}$ in the fifth; with region-by-year effects in their place, $+0.10^{**}$, +0.12 and $+0.45^{*}$; with the comparison firm-years without a rival on record dropped, +0.09, +0.11 and $+1.45^{***}$. At five years, few firm-years are spread over many industry-year and country-year cells, and the estimate depends on which cells hold both exposed and comparison firms. The fifth-year response is positive in every specification, but its size ranges from about half the main estimate to half as much again. We therefore read the response year by year, and mainly through its first three years.

Table 3 builds the specification one block of controls at a time. With only the fixed effects and the lagged change in R&D, the pooled response of log R&D is +0.10. The firm's characteristics and growth leave it at +0.11, the rival-set block raises it to $+0.49^{***}$, and the firm's own deals leave it there. The placebo estimates of log R&D stay close to zero in every column. Those of the inverse hyperbolic sine do not: without the rival-set block,

Table 3. The response of R&D with the controls added one block at a time

Controls	(1) None	(2) + Firm	(3) + Growth	(4) + Rival set	(5) + Own deals
Panel A. Log R&D					
Post-period, $h = 0$ to 5	0.103	0.116	0.110	0.493***	0.491***
	(0.087)	(0.084)	(0.079)	(0.143)	(0.140)
Placebo, $h = -4$ to -2	−0.017	−0.016	−0.019	−0.004	−0.005
	(0.020)	(0.021)	(0.020)	(0.027)	(0.027)
Placebo, $h = -4$ to -3	−0.025	−0.025	−0.028	−0.006	−0.008
	(0.030)	(0.031)	(0.030)	(0.040)	(0.040)
Events at $h = 0$	3,588	3,588	3,588	3,588	3,588
Observations at $h = 0$	7,179	7,015	7,015	7,015	7,015
Panel B. Inverse hyperbolic sine of R&D					
Post-period, $h = 0$ to 5	0.065	0.079	0.085*	0.234***	0.232***
	(0.052)	(0.052)	(0.049)	(0.090)	(0.089)
Placebo, $h = -4$ to -2	−0.032***	−0.031***	−0.032***	−0.011	−0.013
	(0.012)	(0.012)	(0.012)	(0.016)	(0.015)
Placebo, $h = -4$ to -3	−0.048***	−0.047***	−0.048***	−0.017	−0.019
	(0.017)	(0.017)	(0.017)	(0.023)	(0.023)
Events at $h = 0$	4,328	4,328	4,328	4,328	4,328
Observations at $h = 0$	8,762	8,554	8,554	8,554	8,554

Notes: Local-projection difference-in-differences estimates of the change in R&D from the year before the onset of a consolidation episode among the firm's designated rivals, against the firm-years whose rivals made no consolidation in the five years before and none through the horizon. Every column includes two-digit-industry-by-year and country-by-year fixed effects and the lagged change in the outcome. The controls are added one block at a time: the firm's log total assets, log sales, log age, cash over assets and capital expenditure over assets at $t-1$ (firm); the growth of its assets and sales from $t-2$ to $t-1$ (growth); the number of its live designated rivals at $t-1$ and indicators for a year without a rival on record at t and $t-1$ (rival set); indicators for its own consolidations at t, $t-1$ and $t-2$ (own deals). Column (5) is the main specification. Post-period: the pooled average change from $h = 0$ to 5, on the firm-years observed through $h = 5$. Placebo: the pooled average over the placebo years; because the lagged change in the outcome is a control, the estimate at $h = -2$ is zero by construction, and the second row averages the informative years only. Panel A: firm-years with positive R&D; panel B: firm-years that report R&D, zeros included, R&D in millions of dollars. Standard errors clustered by the firm's reporting unit in parentheses. *, **, ***: significance at 10, 5 and 1 per cent.

the exposed firms' R&D falls behind before the onset, by -0.048^{***} over the informative placebo years, and the gap closes once the block enters.

The rival-set block moves the estimate for a mechanical reason. An event requires a recorded rival deal, so no event falls in a year without a rival on record, while 20.5 per cent of the quiet comparison firm-years do. The indicator for such years at t thus gives these comparison firm-years an intercept of their own, and the comparison rests on the firm-years whose rivals are on record and quiet. The unrecorded years are those in which a comparison firm's quiet is least certain: 71.6 per cent of them belong to firms whose rivals had consolidated in an earlier year, against 4.6 per cent of the recorded comparison firm-years. Such firms may still be raising their research spending, which narrows the gap between exposed and comparison firms. Taking the block apart confirms this reading (Appendix Table E2). Added alone to column (3), the indicator at t raises the pooled response to $+0.52^{***}$, the whole effect of the block, while the indicator at $t-1$ alone gives

$+0.16^{*}$ and the number of live rivals alone $+0.18^{*}$.

B. Is the comparison fair?

Before the onset, the research spending of the exposed and the comparison firms moves together. The estimates at $h = -4$ and $h = -3$ are -0.01 and -0.02, with an average of -0.01; for the inverse hyperbolic sine, -0.03, -0.01 and -0.02 (Appendix Table E1).

The two groups differ in their levels (Table 4). Within industry-year and country-year, the exposed firms have larger total assets, by $+0.34^{***}$ log points, higher R&D, by $+0.46^{***}$, and $+1.61^{***}$ more live rivals, and their characteristics jointly predict the event ($F = 22.8$). The long difference removes levels; what the design requires is that the two groups be on the same path, and in growth they are alike. In the year before the onset, the differences in the growth of R&D, total assets and sales are -0.016, -0.014 and -0.016, and they are jointly insignificant ($F = 1.11$, $p = 0.34$).

A stricter test replaces R&D with each of the firm's other characteristics as the outcome of the design (Figure 4). None of them moves before the onset: the pooled placebo estimates are $+0.01$ for log total assets, $+0.00$ for log sales and -0.01 for log employment, and close to zero for the ratios. After the onset the exposed firms grow. By the third year, their total assets are $+0.15^{**}$ log points higher and their employment $+0.36^{***}$, while their sales ($+0.11$), cash, leverage and capital expenditure do not change significantly. The firms that raise their research spending expand their assets and workforce with it, before their sales respond.

Finally, some events have few comparison firms like them. Restricting the comparison to the cells of country and two-digit industry that hold an event in the same year gives a pooled response of $+0.46^{***}$; requiring the same tercile of sales as well gives $+0.55^{***}$, with a pooled placebo estimate of $+0.049^{*}$ (Appendix Figure E2). Against firms like them, the exposed firms respond at least as much.

C. Research spending relative to the firm's size

Because the exposed firms also grow, a rise in their R&D could merely follow their size. R&D over lagged total assets separates the two (Figure 5). It does not move in the first four years after the onset, from $+0.002$ at the onset to -0.017 in the fourth year, and rises in the fifth, by $+0.051^{***}$ against a sample mean of 0.113; the pooled estimate is $+0.036^{**}$, with a pooled placebo estimate of $+0.004$. Research spending first grows with the firm, and by the fifth year it outgrows it.

D. Does the spending become patents?

On the firms matched to patent assignees, patenting does not rise within five years of the onset (Figure 6). The patent families the firm originates are lower in the second

Table 4. The exposed and comparison firm-years before the onset

	Events	Comparison	Standardised difference	Within
Panel A. Firm-years with positive R&D				
Log total assets	6.357	6.040	0.147	0.336*** (0.078)
Log sales	5.855	5.571	0.117	0.321*** (0.081)
Log age	3.355	3.234	0.144	0.081*** (0.028)
Log employment	7.409	7.198	0.095	0.313*** (0.099)
Cash over assets	0.233	0.236	−0.014	−0.007 (0.007)
Capital expenditure over assets	0.042	0.043	−0.025	0.001 (0.001)
Leverage	0.205	0.199	0.030	0.006 (0.007)
R&D (the outcome)	2.386	1.811	0.265	0.465*** (0.083)
R&D growth, $t-2$ to $t-1$	0.073	0.088	−0.024	−0.016 (0.017)
Asset growth, $t-2$ to $t-1$	0.094	0.100	−0.018	−0.014 (0.009)
Sales growth, $t-2$ to $t-1$	0.061	0.072	−0.021	−0.016 (0.013)
Live designated rivals	3.399	1.729	0.455	1.609*** (0.109)
No rival on record	0.414	0.313	0.213	0.063*** (0.016)
Joint test, all			22.8 (<0.001)	
Joint test, growth			1.11 (0.344)	
Events / comparison			3,536 / 3,907	
Panel B. Firm-years reporting R&D				
Log total assets	6.436	6.160	0.128	0.321*** (0.070)
Log sales	5.978	5.663	0.134	0.323*** (0.073)
Log age	3.341	3.224	0.139	0.071*** (0.026)
Log employment	7.518	7.209	0.138	0.310*** (0.086)
Cash over assets	0.211	0.211	0.001	−0.003 (0.006)
Capital expenditure over assets	0.043	0.043	0.002	0.000 (0.001)
Leverage	0.227	0.225	0.007	0.006 (0.007)
R&D (the outcome)	2.666	2.216	0.220	0.372*** (0.065)
R&D growth, $t-2$ to $t-1$	0.060	0.059	0.003	0.006 (0.009)
Asset growth, $t-2$ to $t-1$	0.091	0.095	−0.012	−0.007 (0.008)
Sales growth, $t-2$ to $t-1$	0.060	0.072	−0.024	−0.005 (0.012)
Live designated rivals	3.546	1.624	0.527	1.730*** (0.099)
No rival on record	0.390	0.339	0.107	0.030* (0.015)
Joint test, all			28.3 (<0.001)	
Joint test, growth			0.46 (0.712)	
Events / comparison			4,260 / 4,783	

Notes: The events and the comparison firm-years of the main specification in the year of the onset, on the firm's characteristics in the year before; the firm-years with positive R&D (Panel A) are the sample of log R&D, those reporting R&D, zeros included (Panel B), the sample of its inverse hyperbolic sine, and R&D (the outcome) is measured accordingly. Events and Comparison: means. Standardised difference: the difference in means over the square root of the average variance. Within: the difference within industry-year and country-year, the coefficient of an event indicator in a regression of the characteristic on it with two-digit-industry-by-year and country-by-year fixed effects, standard error clustered by the firm's reporting unit in parentheses. Joint tests: the event indicator regressed on all the characteristics with the same fixed effects; the F statistic, with its p-value in parentheses, for all the characteristics and for the three growth rates. Log employment is reported by fewer firms. *, **, ***: significance at 10, 5 and 1 per cent.

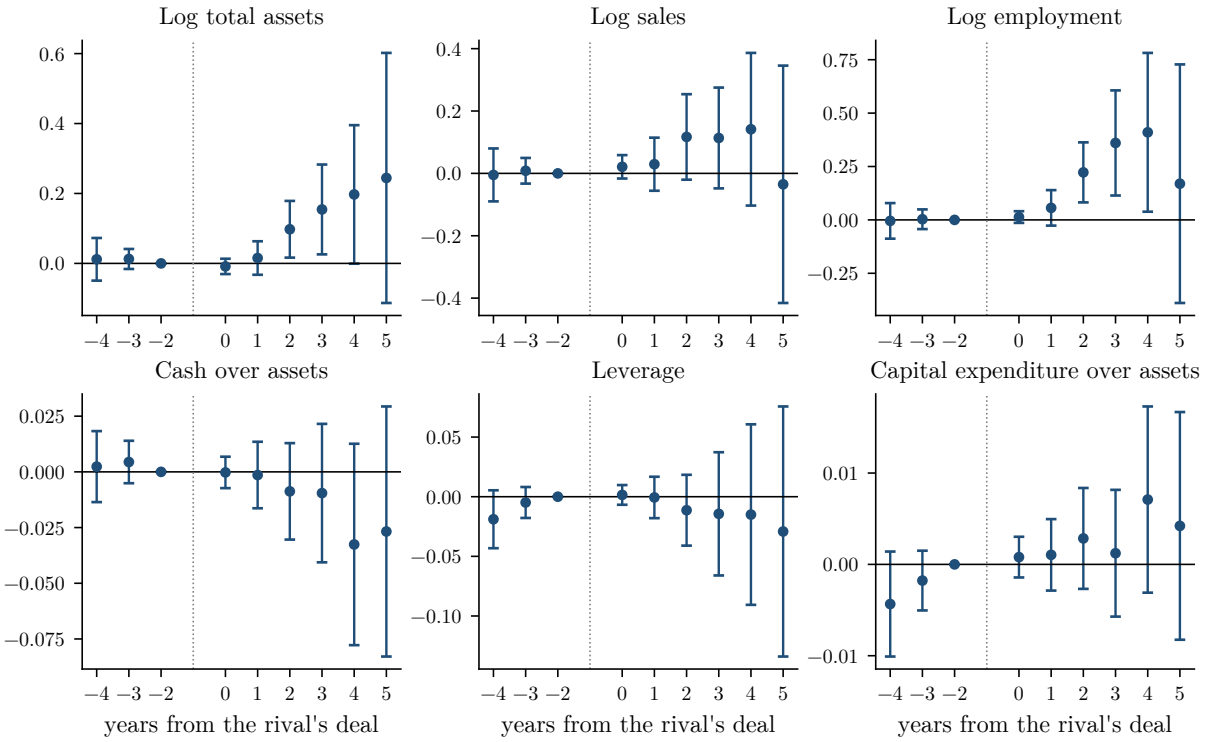


Figure 4. The firm's characteristics around the onset

Notes: Each panel replaces the change in R&D in the main specification by the change in one of the firm's characteristics from the year before the onset to year h, on the firm-years of the log R&D sample, with the same comparison group, fixed effects and controls; the characteristic's own lagged level and growth are among the controls when it is one of them. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

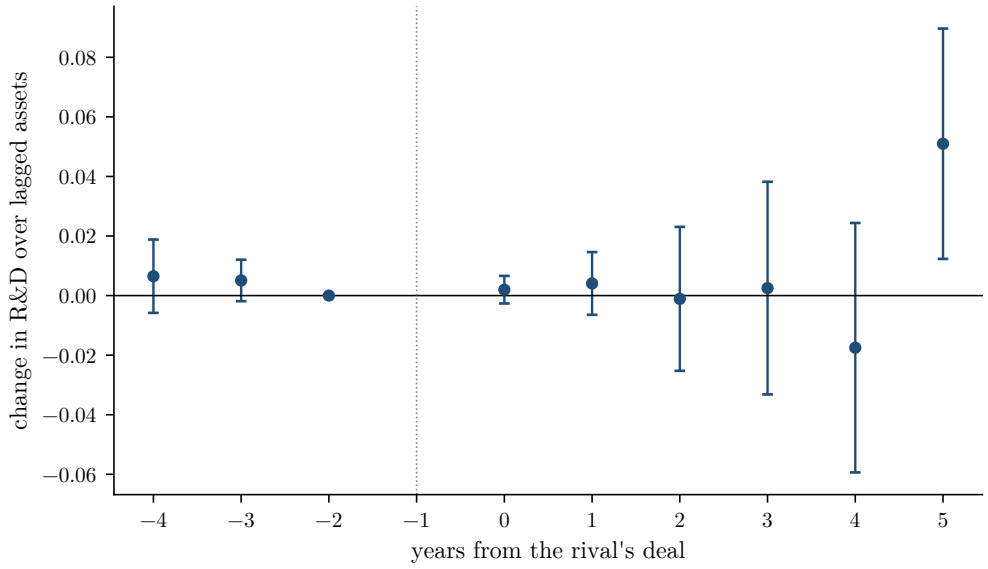


Figure 5. The response of R&D over lagged total assets

Notes: R&D in the year over the firm's total assets of the year before, winsorised, as the outcome of the main specification (3,138 events at $h = 0$; firm-years that report R&D, zeros included). Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference, and the estimate at $h = -2$ is zero by construction.

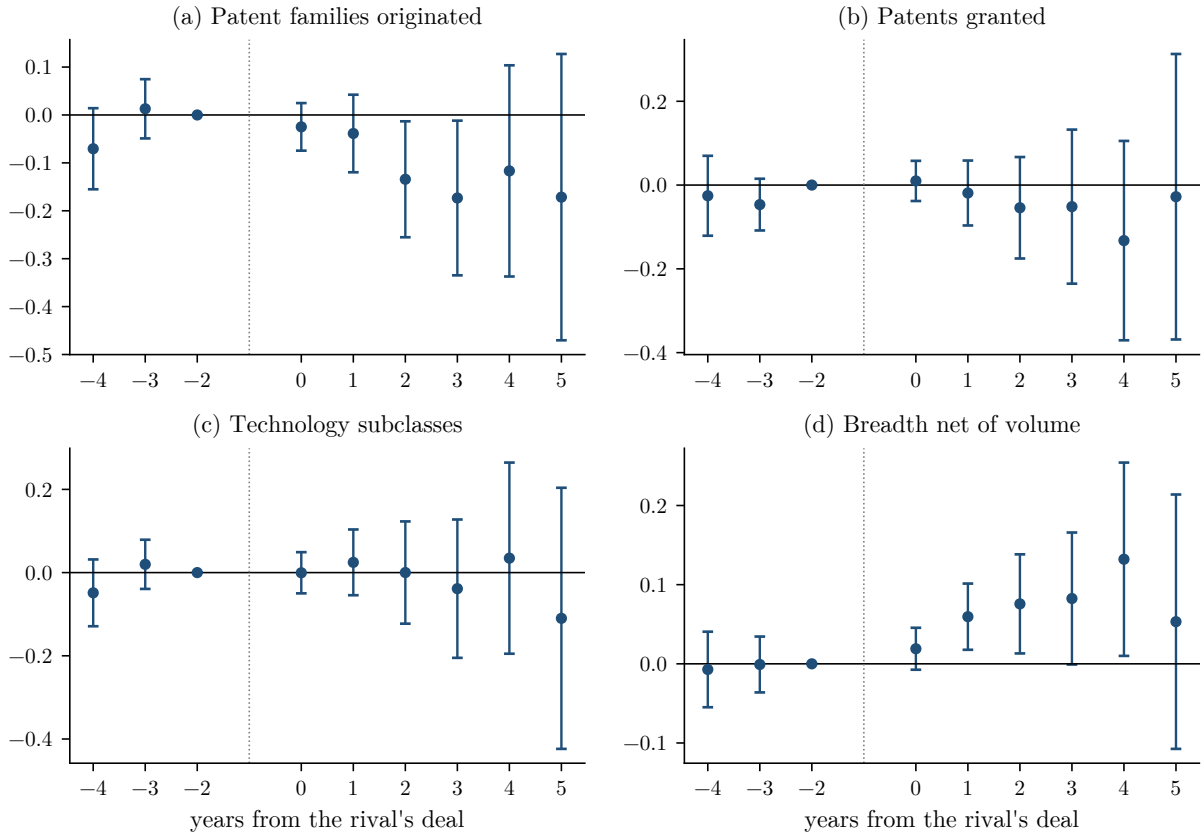


Figure 6. The response of patenting

Notes: The main specification on the firms matched to patent assignees, with the inverse hyperbolic sine of the patent families the firm originates, by priority year (panel a), of its patents granted, by filing year (panel b), and of the distinct technology subclasses of its families (panel c); panel (d) is the breadth of its patenting net of volume, the inverse hyperbolic sine of the links between its families and their subclasses, each family counted once in each of its subclasses, minus that of the families of panel (a). 5,004 events at $h = 0$. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

and third years, by -0.13^{**} and -0.17^{**}, and indistinguishable from zero afterwards; its granted patents and the technology subclasses of its families do not move. Pooled over the six years, the estimates are -0.08 for families, $+0.05$ for granted patents and -0.03 for subclasses, with flat placebo years.

What changes is how widely the firm patents. With as many subclasses and fewer families, each family spans more technology classes. We measure this breadth net of volume: the inverse hyperbolic sine of the links between the firm's families and their technology subclasses, each family counted once in each of its subclasses, minus that of the families. It rises in each of the first four years, from $+0.06^{***}$ in the first to $+0.13^{**}$ in the fourth, with flat placebo years, although its pooled estimate is imprecise ($+0.08$).[16] Within five years, the exposed firms spend more on research and spread their patenting over more technology classes, without patenting more.

[16] A family without a CPC code adds no link. Net of the families that carry a code, breadth is $+0.04^{***}$, $+0.05^{**}$, $+0.04$ and $+0.08^{**}$ in the first four years, with flat placebo years. The two measures differ only through the share of the firm's families that carry a code.

The estimates for patent counts turn positive seven and eight years after the onset, but on samples that the long horizon selects: an event reaches the seventh year only if it is old enough and no new onset intervenes, and its comparison firm-years only if their rivals make no deal for thirteen years. We read these horizons as consistent with a long lag between research spending and patents, not as an estimate of it (Appendix F).

IV. Whose consolidation matters?

A. The direction of attention

A firm's competitors fall into three disjoint sets: the rivals it names that do not name it back, the mutual rivals, and the firms that name it without being named back. If the response runs through attention, it should follow the consolidations of the first set and not those of the third. If it runs through the market the two firms share, the direction of naming should not matter. We estimate the main design on each set in turn, with the consolidations of the other two sets in the controls.[17]

The response follows the direction of attention (Figure 7). Pooled over the six years, log R&D rises by $+0.37^{***}$ after a consolidation by a rival the firm names one way, by $+0.17^{***}$ after one by a mutual rival, and by $+0.08$ after one by a firm that names the firm without being named back. The inverse hyperbolic sine gives the same ranking, $+0.19^{***}$, $+0.10^{**}$ and $+0.06$, and R&D over lagged assets separates the directions by sign: it does not fall after a consolidation by a rival the firm watches ($+0.014$) and falls after one by a firm that watches it (-0.029^{***}). Year by year, the response to the firms that name the firm stays at zero, from $+0.00$ in the year of the deal to $+0.02$ in the fifth year, while the responses to the rivals the firm names rise from the first year on. These two are alike through the third year, $+0.15^{*}$ for rivals named one way and $+0.11^{*}$ for mutual rivals, and the larger pooled response to rivals named one way comes from the fourth and fifth years ($+0.24^{**}$ and $+0.62^{***}$, against $+0.19^{**}$ and $+0.22^{**}$). What separates the directions is whether the firm watches the consolidating firm, not whether it is watched back.[18]

B. Attention, not the size of the rival

Because firms name larger rivals, the direction of naming and the size of the consolidating firm go together. At the events, the consolidating firm has higher sales than the focal firm

[17] Each estimate in this section is the main specification on one group of events. We report pooled estimates, given with their placebo periods in Table 5 and Appendix Table G1, and check each comparison year by year, since the pooled estimates weigh the fifth year, the least precise.

[18] Not every designated rival is a source firm, so a rival named one way may not name the firm back only because FactSet records no competitors for it. When every consolidating rival named one way names competitors of its own in the year, so that we observe that it does not name the firm, log R&D rises by $+0.40^{***}$ (1,280 events); when at least one names none, it rises by $+0.36^{***}$ (2,376 events). The inverse hyperbolic sine gives $+0.22^{***}$ and $+0.18^{***}$, and the placebo years are flat in both groups.

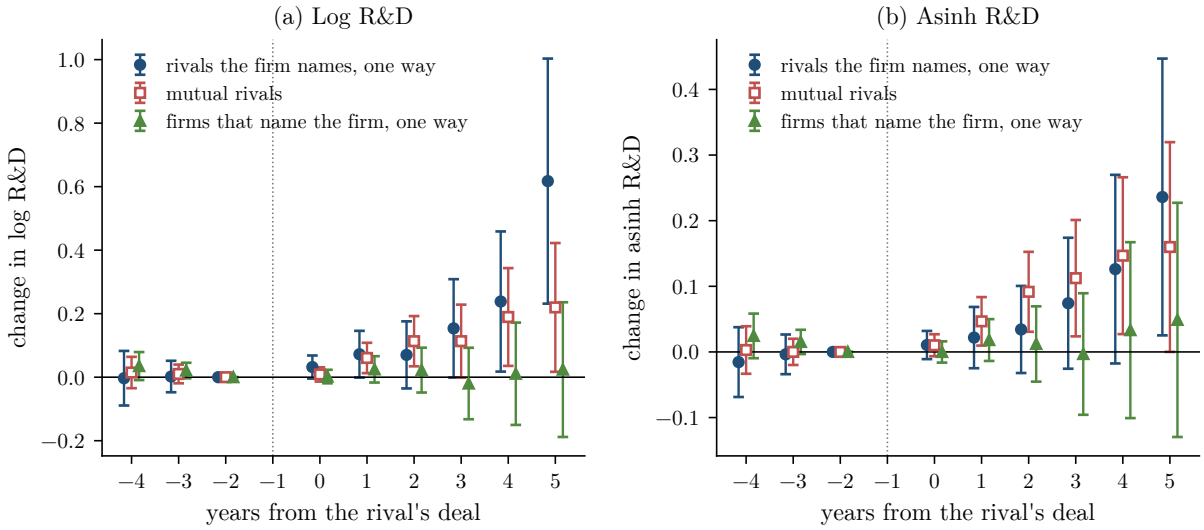


Figure 7. The response by the direction of naming

Notes: Three disjoint sets of competitors: the rivals the firm names that do not name it back, the mutual rivals, and the firms that name the firm without being named back. For each set, the event is a consolidation by a member after a year with none, the comparison group the firm-years with no such deal in the five years before and none through the horizon, the set's own size and reporting indicators replace those of the designated set among the controls, and the consolidations of the other two sets enter the controls at t, $t-1$ and $t-2$; otherwise the specification is the main one. Events at $h=0$ for log R&D: 3,725, 3,777, 4,166. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

in 96 per cent of cases for rivals named one way, in 74 per cent for mutual rivals and in 49 per cent for firms that name the firm. We therefore compare the directions at a given size (Figure 8). When the largest consolidating firm has more than ten times the focal firm's sales, log R&D rises by $+0.42^{***}$ if the firm names it and by -0.01 if it names the firm. The contrast holds in every year: after a much larger rival the firm names consolidates, its R&D rises from the year of the deal, by $+0.05^{**}$, to $+0.19^{**}$ in the third year, while after a much larger firm that names it consolidates, its R&D does not move (-0.00 in the third year). When the consolidating firm has one to ten times the firm's sales, the responses are smaller and do not follow the direction of naming: $+0.16$ for a rival named one way, $+0.15^{**}$ for a mutual rival and $+0.12^{*}$ for a firm that names the firm, the last with a positive placebo period. At these sizes the data cannot separate the directions. A deal by a firm closer to the focal firm's size may be a smaller event, and one-way naming between firms of similar size may say less about who watches whom.

C. The strength of attention

If the response follows attention, it should be weaker where the attention is new. We group the events by how long the firm had named its longest-standing consolidating rival when that rival dealt.[19] When the oldest tie dates from the year of the deal or the year

[19] The age of a tie is truncated by the start of the firm's records, so new ties are more frequent among the firms covered more recently; the groups differ in the firms they contain as well as in the age of the tie.

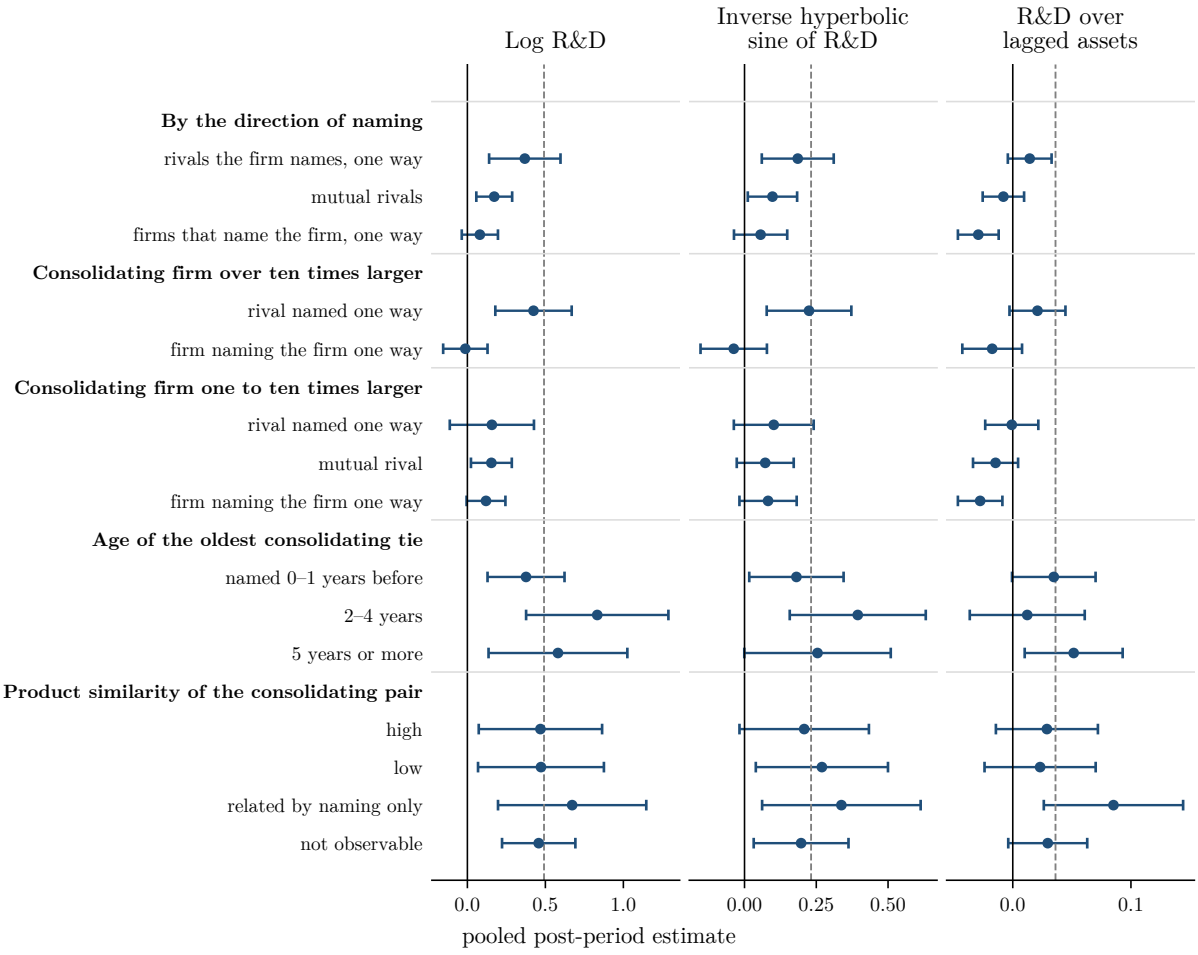


Figure 8. Whose consolidation matters

Notes: The pooled post-period estimate ($h = 0$ to 5) with its 95 per cent confidence interval, by the direction of naming, by the direction at a given size of the consolidating firm relative to the firm, by the age of the firm's oldest tie to a consolidating rival, and by the product similarity of the closest consolidating pair in the text-based network; the dashed line is the main estimate. Appendix Table G1 gives the estimates with their placebo periods and events.

before, the pooled response of log R&D is $+0.38^{***}$; when it is five years or older, $+0.58^{**}$, with flat placebo years in both groups. The inverse hyperbolic sine gives $+0.18^{**}$ and $+0.25^{*}$, and R&D over lagged assets rises most after the consolidations of the oldest ties ($+0.052^{**}$). The events whose oldest tie is two to four years old show the largest estimate, $+0.83^{***}$, but their research spending was already falling behind before the onset (a pooled placebo estimate of -0.062^{*}), and we do not read it. Year by year, the two clean groups do not differ. The response to new ties is significant from the year of the deal ($+0.05^{**}$); that to older ties is of similar size but less precise ($+0.32^{*}$ against $+0.23^{**}$ in the third year), and its larger pooled estimate comes from the fifth year. The age of the tie does not order the response.

D. Attention, not product overlap

Firms may name the competitors closest to them in product space, and the response could follow product overlap rather than attention. Two facts speak against this. The mutual rivals are the closest pairs in product space, related in the text-based network in 91 per cent of the observable cases against 65 per cent for pairs named one way, yet the response to their consolidations is the smaller one. And grouped by the largest product similarity between the firm and a consolidating rival, the pooled responses are alike: $+0.47^{**}$ for the most similar pairs, $+0.47^{**}$ for the less similar, $+0.67^{***}$ for named rivals that the network does not relate to the firm, and $+0.46^{***}$ for the events outside its universe. Year by year, the most similar pairs respond earlier, $+0.23^{***}$ against $+0.05$ in the first year, but these groups are small (308 and 287 events). Product overlap does not account for the response to named rivals, although it may speed it.

E. Attention, not the structure of the market

If the firm responded to the change a consolidation brings to the structure of its market, the response would grow with the share of the rival set that consolidates, with the size of the acquiring rival's step, with deals that join two of its rivals and with the rival's proximity in industry and in location, and it would depend on whether the rival buys or is bought and on whether the industry is in a merger wave. It does none of these (Table 5). The pooled response of log R&D is $+0.50^{***}$ when at least half of the firm's live rivals consolidate in the year and $+0.49^{***}$ when fewer do; $+0.52^{***}$ after an acquisition that is large relative to the acquiring rival and $+0.46^{***}$ after a smaller one; $+0.46$ when one rival acquires another and $+0.49^{***}$ after any other rival deal; $+0.46^{***}$ when the rival is acquired and $+0.49^{***}$ when it acquires. It is $+0.47^{***}$ for a consolidating rival in the firm's four-digit industry, $+0.46^{***}$ for one in its two-digit industry only and $+0.59^{***}$ for one in another industry. Location does not matter either: the response is $+0.45^{***}$ when the nearest consolidating rival has its headquarters within 622 km of the firm's, the median

distance, and $+0.53^{***}$ when it is farther; it is $+0.47^{***}$ when a consolidating rival is in the firm's country and $+0.54^{***}$ when none is. It is $+0.41^{**}$ for a deal during a merger wave in the firm's four-digit industry, the top quarter of the industry's years of worldwide deal activity, against $+0.52^{***}$ outside one. Year by year, no split shows one group's response consistently above the other's.

The sharpest contrast replaces the rivals a firm names with those an industry definition would assign it: its four-digit industry peers in the panel, with everything else as in the main specification (Table 6). Year by year, the firm's R&D falls slightly in the year of the peer's deal (-0.06^{***}) and does not change in the three years after it (-0.00 in the third), whereas it rises from the year of a named rival's deal. Its estimates turn positive in the fourth and fifth years ($+0.16^{*}$ and $+0.15$), on few firm-years. The pooled estimate, which rests on the firm-years observed for six years, is $+0.03$. Because 87 per cent of the firm-years have a peer in a deal, quiet years are rare: 41 per cent of the comparison firm-years belong to firms without an industry peer in the panel, and 66 per cent have a designated rival's consolidation in the same year. The same pattern, flat for three years and higher afterwards, appears when the designated rivals' deals at t, $t-1$ and $t-2$ enter the controls (-0.04 in the third year and $+0.16$ in the fourth) and when the comparison keeps only firms with an industry peer ($+0.02$ and $+0.41^{**}$). Only 17 per cent of the industry onsets involve a peer the firm names, and on those the estimate is imprecise (-0.08). An industry definition thus misses the response that follows a named rival's consolidation in the first years; the data cannot establish whether a peer's deal produces a delayed response of its own. Random rival sets drawn from the firm's industry, which keep the size of the real set and its quiet years, give no rise at all (Section VI.F).

F. What the evidence leaves

Explanations that run through the structure of the market predict that the response does not depend on the direction of naming, since a merger changes the market of every firm in it, and that it grows with the change in structure: with the share of the market that consolidates, with horizontal deals and with the proximity of the merging firms. This holds for the incentive to escape competition by innovating (Aghion et al. 2005), for the demand a merger diverts to the firms outside it (Federico, Langus and Valletti 2018; Motta and Tarantino 2021), and for the rents and the quiet life that less competition allows (Hicks 1935; Arrow 1962). The data reject both predictions. The response follows whom the firm watches, and it is clearest when a much larger rival the firm watches consolidates. This contrasts with the evidence on rivals defined by industry: among the rivals of US horizontal mergers, those much smaller than the merging firms cut their R&D (Li 2024), as the escape-competition account predicts for firms far behind.

Three readings fit this pattern, and the data do not separate them. The firm may keep up with a rival that has become stronger: a rival's move creates competitive tension only

Table 5. The response by how the consolidation changes the market

	Log R&D			Inverse hyperbolic sine of R&D		
	After	Before	Events	After	Before	Events
Named rivals (main)	0.491***	−0.005	3,588	0.232***	−0.013	4,328
	(0.140)	(0.027)		(0.089)	(0.015)	
Share of the rival set in deals						
at least half	0.496***	0.000	1,098	0.255**	−0.015	1,323
	(0.167)	(0.033)		(0.101)	(0.020)	
fewer than half	0.489***	−0.007	2,490	0.221**	−0.012	3,005
	(0.146)	(0.031)		(0.093)	(0.018)	
Size of the acquiring rival's step						
above the median	0.524***	−0.005	1,198	0.268***	−0.011	1,471
	(0.173)	(0.034)		(0.102)	(0.020)	
below the median	0.462***	0.009	1,226	0.223**	−0.002	1,427
	(0.144)	(0.036)		(0.087)	(0.023)	
no valued acquisition	0.481***	−0.020	1,164	0.184*	−0.024	1,430
	(0.162)	(0.036)		(0.111)	(0.020)	
Horizontal deal						
a rival buys another rival	0.461	−0.072	124	0.277	−0.053	175
	(0.291)	(0.082)		(0.181)	(0.041)	
any other rival deal	0.493***	−0.003	3,464	0.229***	−0.011	4,153
	(0.140)	(0.027)		(0.087)	(0.016)	
Role of the rival						
rival acquired	0.460***	−0.065	621	0.230*	−0.049**	812
	(0.176)	(0.044)		(0.123)	(0.025)	
rival acquiring	0.488***	−0.005	3,428	0.232***	−0.010	4,114
	(0.142)	(0.027)		(0.089)	(0.016)	
Industry of the consolidating rival						
same four-digit industry	0.474***	−0.027	1,386	0.222**	−0.025	1,777
	(0.159)	(0.035)		(0.095)	(0.020)	
same two-digit only	0.458***	0.011	1,005	0.236**	−0.005	1,110
	(0.167)	(0.036)		(0.105)	(0.022)	
another industry	0.592***	0.020	1,116	0.263**	0.005	1,337
	(0.165)	(0.035)		(0.104)	(0.022)	
Location of the consolidating rival						
nearest within the median distance	0.449***	−0.020	1,298	0.168*	−0.025	1,571
	(0.155)	(0.035)		(0.096)	(0.020)	
nearest beyond it	0.526***	0.010	2,207	0.281***	−0.002	2,651
	(0.147)	(0.030)		(0.093)	(0.017)	
one in the firm's country	0.473***	−0.009	1,757	0.216**	−0.016	2,367
	(0.160)	(0.032)		(0.097)	(0.018)	
none in the firm's country	0.544***	0.008	1,750	0.282***	−0.002	1,857
	(0.139)	(0.033)		(0.097)	(0.021)	
Industry merger wave						
inside a wave	0.413**	0.009	1,105	0.245**	0.002	1,362
	(0.173)	(0.036)		(0.103)	(0.021)	
outside a wave	0.519***	−0.011	2,483	0.227**	−0.019	2,966
	(0.146)	(0.029)		(0.093)	(0.017)	

Notes: The main specification on subsets of the events, against the main comparison group (Section IV.E defines each split). After: the pooled estimate over $h = 0$ to 5; Before: over the placebo years $h = -4$ to -2; standard errors clustered by the firm's reporting unit in parentheses; Events: at $h = 0$. *, **, ***: significance at 10, 5 and 1 per cent.

Table 6. The response with industry peers as the rival set

	Log R&D			Inverse hyperbolic sine of R&D		
	After	Before	Events	After	Before	Events
Named rivals (main)	0.491***	−0.005	3,588	0.232***	−0.013	4,328
	(0.140)	(0.027)		(0.089)	(0.015)	
Industry peers as the rival set						
all industry onsets	0.032	0.002	1,779	0.072	0.004	2,200
	(0.082)	(0.025)		(0.047)	(0.017)	
a named rival among the peers	−0.081	0.033	297	0.077	0.019	389
	(0.204)	(0.036)		(0.110)	(0.023)	
designated rivals'	0.057	0.002	1,779	0.115*	0.005	2,200
deals controlled	(0.186)	(0.025)		(0.061)	(0.017)	
comparisons with an industry peer	−0.071	0.019	1,779	0.089	0.028	2,200
	(0.282)	(0.028)		(0.072)	(0.021)	

Notes: The main specification with the firm's four-digit industry peers in the panel as its rival set: the event is a peer's consolidation after a year without one, and the comparison firm-years are those whose peers make no deal in the five years before and none through the horizon. The second row keeps the industry onsets in which a dealing peer is also a rival the firm names; the third adds the designated rivals' consolidations at t, $t-1$ and $t-2$ to the controls; the fourth keeps only the comparison firm-years of firms with at least one industry peer in the panel. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

for the firms aware of it (Chen, Su and Tsai 2007). The firm may learn from the rival's deal about where the market is going (Lieberman and Asaba 2006). Or the firm may search harder because it falls further behind a peer it compares itself with, the problemistic search of the behavioural theory of the firm, in which research spending rises when performance falls short of the reference group's (Cyert and March 1963; Greve 2003). One result bears on the second reading: deals that are announced and later withdrawn are followed by no significant change in research spending (Section VI.B), so the news of an attempted consolidation is not enough.

V. Which firms respond?

A. Innovation assets

We split the events by the firm's characteristics before the onset (Figures 9 and 10).[20] The response appears in every group. We begin with the firm's research capital: its R&D of the five years before the onset, depreciated at 15 per cent a year, over its total assets in the year before. The pooled response of log R&D is +0.34** in the lowest third, +0.51***

[20] Each group's events are compared with the main comparison group under the main specification, so the groups differ only in the events they contain. The confidence intervals of the groups overlap, and we read their differences as tendencies, year by year, since several groups rank differently in the pooled estimates, which weigh the fifth year, than in the years before it.

in the middle third and +0.54*** in the highest, but this ranking comes from the fifth year and is reversed before it. The firms with the least research capital raise their R&D from the year of the onset, by +0.07**, then by +0.19*** a year later and +0.28** by the third year; the firms with the most show no rise in the first two years (−0.01 in the first) and an imprecise one in the third (+0.15).

The firm's other intangible assets point the same way. Under most accounting standards, the intangible assets on a balance sheet are those a firm has bought, through acquisitions or purchases of patents and brands, rather than built through its own research. The firms with the most of them respond least: the pooled response is +0.40** in the highest third of intangible assets over total assets, against +0.50*** in the middle and +0.65*** in the lowest, and +0.12 by the third year against +0.32** in the lowest third. Among the firms matched to patent assignees, those that originated no patent family in the five years before the onset respond as much as those that patent most: +0.65*** without patents, against +0.46*** and +0.47*** with a patent stock below or above the median, with broadly similar yearly estimates.

The response is thus not concentrated among the firms with the largest research capital. It comes earliest where research capital is modest, is weakest where acquired intangible assets are largest, and does not require a record of patenting.

B. Size and financial resources

Smaller firms respond more. By thirds of total assets, the pooled response of log R&D is +0.63*** among the smallest firms, +0.49*** in the middle and +0.30* among the largest; for the inverse hyperbolic sine, +0.29**, +0.26*** and +0.15. The gradient holds in every year, from the year of the onset, +0.09*** among the smallest firms against +0.00 among the largest, to the third, +0.31** against +0.11. This fits the response to much larger rivals in Section IV.B: the smaller the firm, the larger the rivals it watches are relative to it.

Cash does not order the response: by thirds of cash over total assets, the pooled response is +0.44**, +0.34** and +0.64***, and no third leads the others consistently year by year. Debt may slow it. By thirds of leverage, the pooled response is +0.59***, +0.43*** and +0.39**. The least indebted firms raise their R&D from the year of the onset, by +0.08***, and by +0.17*** a year later, while the most indebted show no significant rise before the fifth year (+0.05 in the first year, +0.17 in the third). Within the range of firms in the sample, financing does not prevent the response, but debt may delay it. The direction matches that among the rivals of US horizontal mergers, where financially constrained firms cut their R&D and unconstrained firms raise it (Li 2024).

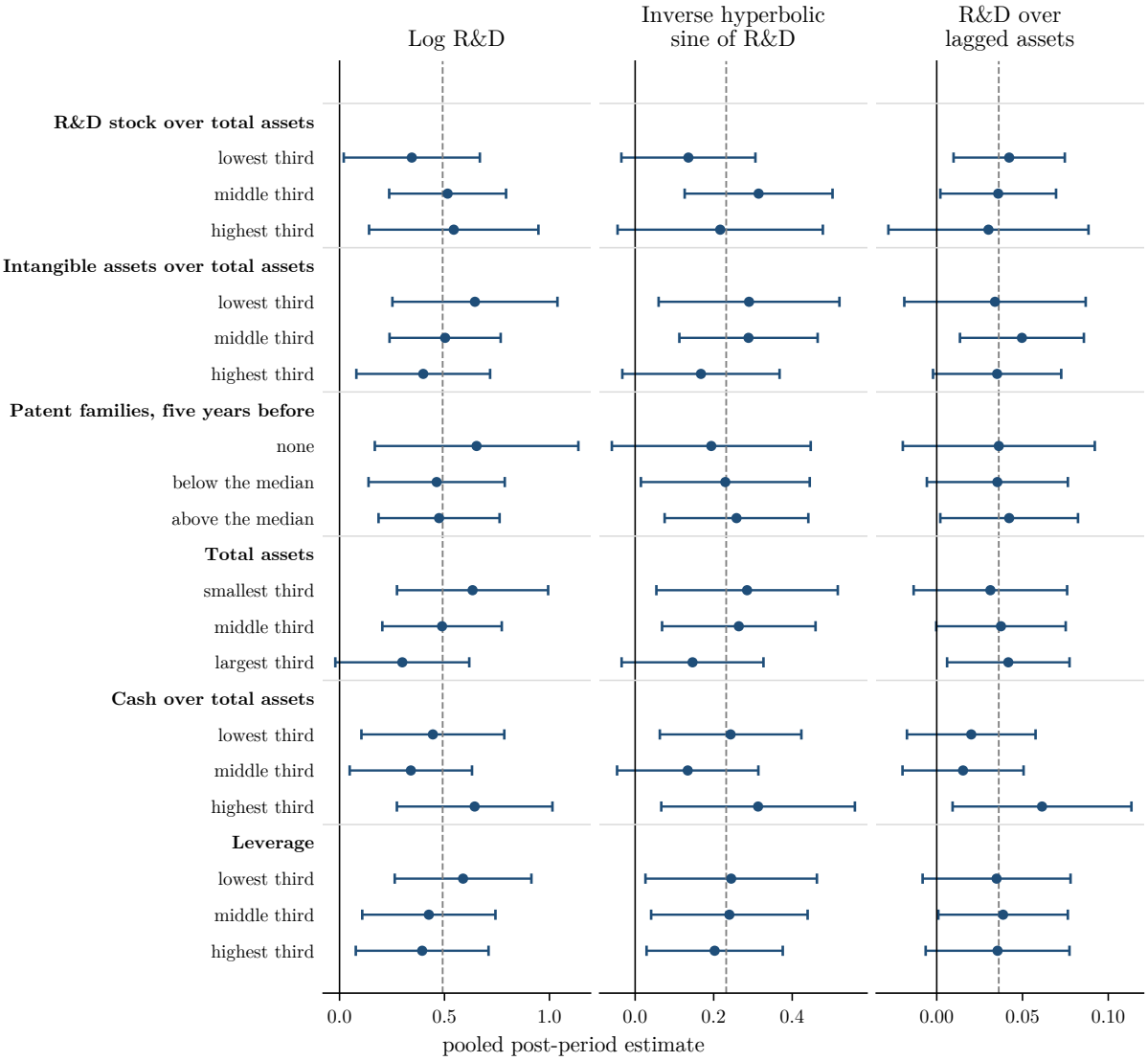


Figure 9. Which firms respond

Notes: The pooled post-period estimate ($h = 0$ to 5) with its 95 per cent confidence interval, by the firm's innovation assets, size and financial resources before the onset; the dashed line is the main estimate. Each row is the main specification on the events of one group, against the main comparison group. R&D stock: R&D of the five years before, depreciated at 15 per cent a year, over total assets at $t - 1$; intangible assets, total assets, cash and leverage at $t - 1$, in terciles of the estimation sample; patent families originated in the five years before, on the firms matched to patent assignees. Appendix Tables G2 and G3 give the estimates with their placebo periods and events.

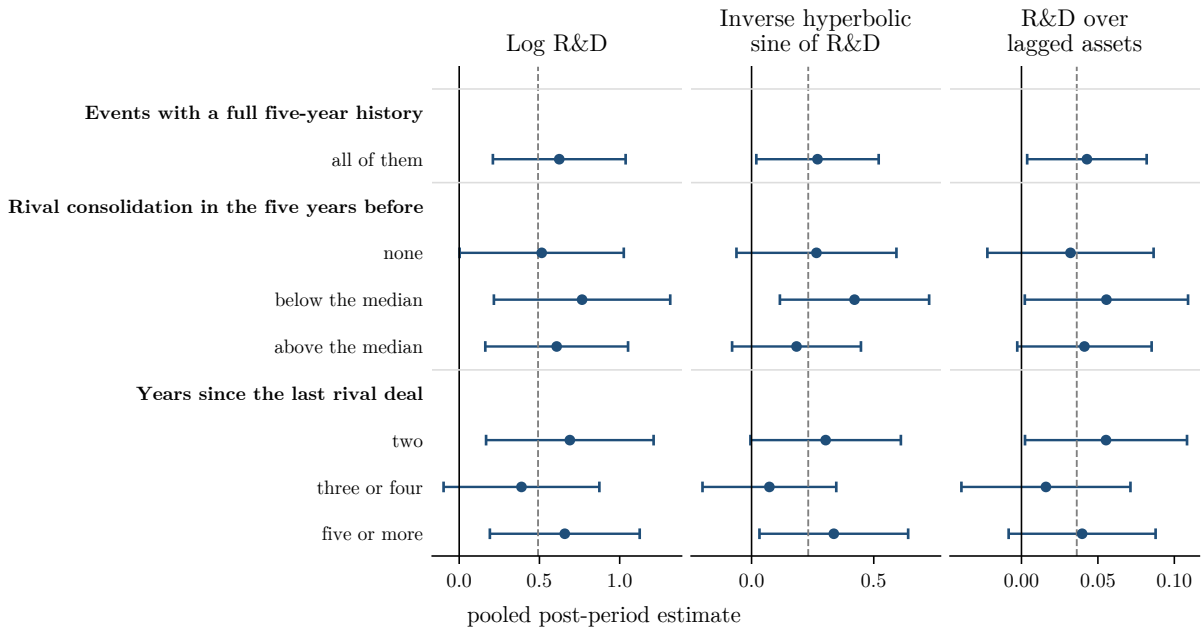


Figure 10. Prior exposure

Notes: The pooled post-period estimate ($h = 0$ to 5) with its 95 per cent confidence interval on the events whose rivals' deals are observed in each of the five years before the onset, by the stock of rival consolidation in those years and by the years since the last rival deal; the dashed line is the main estimate. Each row is the main specification on the events of one group, against the main comparison group. Appendix Table G4 gives the estimates with their placebo periods and events.

C. Prior exposure

A firm whose rivals consolidated recently or often might respond differently to a new onset: its attention may be sharper, or its earlier response may still be under way. We form these groups on the events whose rivals' deals are observed in each of the five years before the onset, since a shorter history cannot distinguish quiet rivals from records that had not yet begun. On these events, the pooled response of log R&D is $+0.62^{***}$ (1,273 events), but year by year it is positive and imprecise until the fifth year (+0.07 in the first year and +0.27 in the third).

By the stock of earlier rival consolidation, the sum over $t-5$ to $t-2$ of the share of the firm's designated rivals that consolidated in each year, the pooled response is $+0.51^{**}$ when the rivals made no deal, $+0.77^{***}$ for a stock below the median and $+0.61^{***}$ for one above it. By the years since the last rival deal, it is $+0.69^{***}$ for two years, +0.39 for three or four and $+0.66^{***}$ for five or more. Year by year, few of these groups show a significant response before the fifth year, and none stands out.

Two features limit what these groups can say. For a firm exposed before, the estimate includes any response to the earlier deals that is still under way, since the comparison firms have seen no rival deal for five years; the groups measure the total change in research spending after the onset, not the increment the new onset adds. And the groups are small, so their differences, including the larger pooled response at an intermediate stock of earlier consolidation, lie within their confidence intervals. These data cannot settle whether the

response depends on how often and how recently a firm has seen its rivals consolidate, as a dynamic version of the relation between competition and innovation would suggest (Aghion et al. 2005).

VI. What else could explain the response?

A. Other deals at the same time

A rival's consolidation rarely comes alone. The firm itself may be acquiring or being acquired, and its suppliers, customers, partners, the rivals of its rivals or its industry as a whole may be consolidating at the same time. We test each channel in two ways: by adding its deals to the controls, and by taking them as the event. Figure 11 gathers the pooled response under every variant of the main specification; none moves it far from the main estimate.

The main specification already controls for the firm's own deals at t, $t-1$ and $t-2$. Leaving out the indicator at t, which could reflect a quick reaction to the rival's deal, gives $+0.50^{***}$. Removing every firm-year with an own deal in that window, from the events and the comparison group alike, gives $+0.56^{***}$ on the 2,300 events that remain, and $+0.28^{**}$ for the inverse hyperbolic sine. With the deals of the firm's suppliers and customers in the controls, the response of log R&D is $+0.37^{***}$; with those of its partners, $+0.49^{***}$; with those of the rivals of its rivals, $+0.56^{***}$; and with its industry's merger wave, $+0.47^{***}$. With all of them at once, it is $+0.37^{*}$ for the log and $+0.11$ for the inverse hyperbolic sine. The channels are correlated with one another and with the rivals' deals, so this last specification asks the rival's deal to explain what remains after every related deal at t, $t-1$ and $t-2$; the log estimate keeps three quarters of its size.

Taken as the event, the deals of the firm's suppliers, customers and partners produce no rise (Table 7). Their point estimates are negative, but we do not read them: the comparison firms of these designs are quiet in the channel and can still see a designated rival's consolidation within the horizon, which raises their spending and pulls the estimates down. Deals among the rivals of its rivals are followed by $+0.30^{**}$, and the onset of an industry merger wave by $+0.44^{***}$. These events exclude a designated rival's deal in the year of the event but not in the years after. Keeping only the events followed by no designated rival's deal through $t+5$, the wave is followed by $+0.08$ and a deal among the rivals of rivals by $+0.41$.[21] On its own, an industry's merger wave is not followed by higher research spending; the rise after a wave comes from the deals of the rivals the firm names.

[21] The restriction conditions on the future and is a check rather than a design: it keeps the waves that were not followed by a designated rival's consolidation, which may be those that mattered least for the firm's rivals.

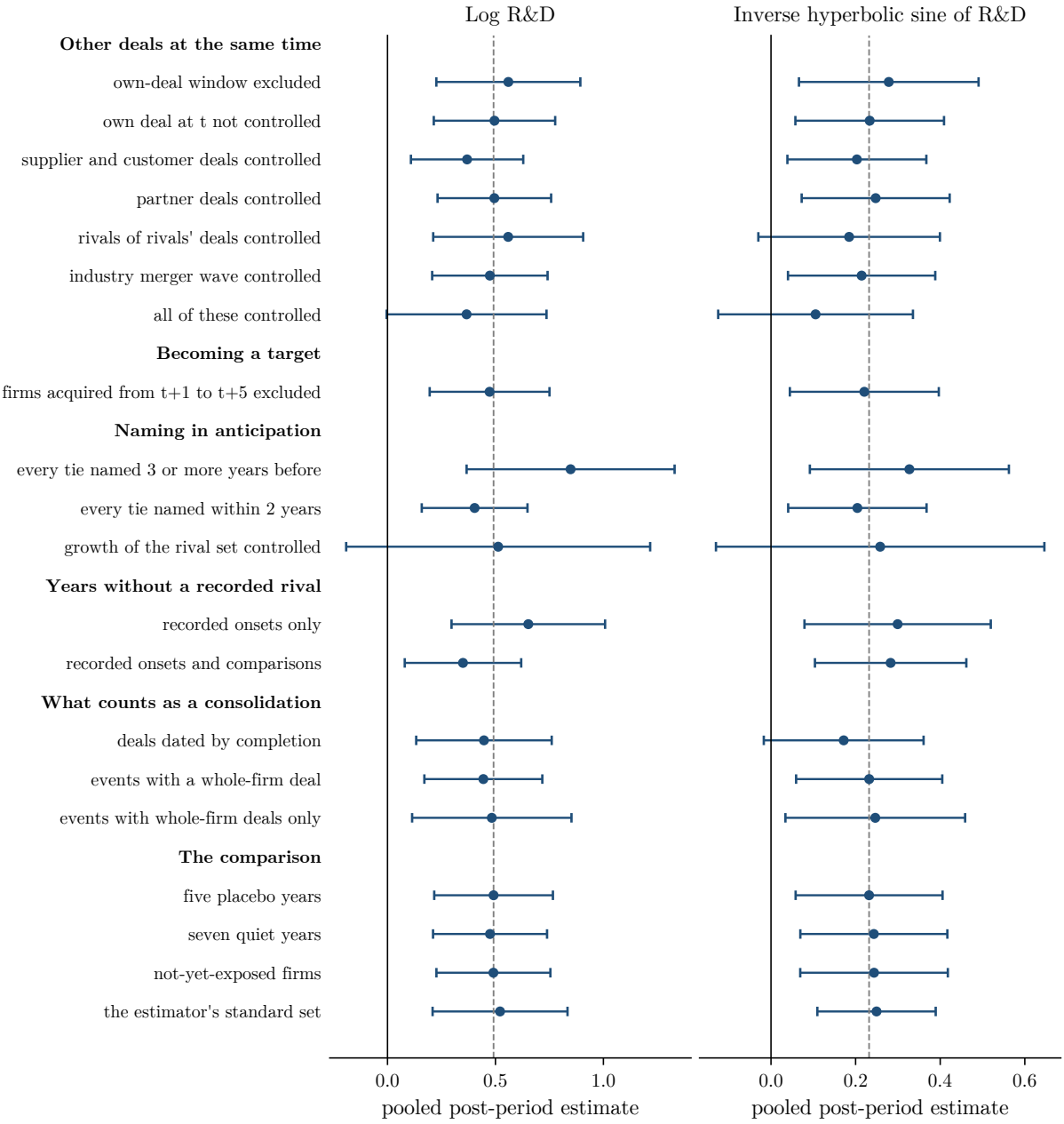


Figure 11. The response across the alternative explanations

Notes: The pooled post-period estimate ($h = 0$ to 5) of each variant of the main specification with its 95 per cent confidence interval; the dashed line is the main estimate. Standard errors clustered by the firm's reporting unit. Appendix Tables H1 and H2 give the estimates with their placebo periods and events.

Table 7. Other deals as the event

	Log R&D			Inverse hyperbolic sine of R&D		
	After	Before	Events	After	Before	Events
A deal elsewhere around the firm as the event						
supplier or	−0.726*	0.024	1,120	−0.111	0.008	1,348
customer deal	(0.434)	(0.050)		(0.102)	(0.025)	
partner deal	−0.133	−0.032	1,021	0.012	−0.015	1,242
	(0.121)	(0.045)		(0.055)	(0.020)	
deal among	0.303**	0.021	1,013	0.126*	−0.012	1,236
rivals of rivals	(0.122)	(0.046)		(0.071)	(0.028)	
industry merger wave	0.444***	0.027	1,449	0.325***	−0.001	1,805
	(0.155)	(0.033)		(0.110)	(0.022)	
Followed by no designated rival deal to $t+5$						
deal among	0.410	0.052	386	0.066	0.021	430
rivals of rivals	(0.292)	(0.050)		(0.155)	(0.033)	
industry merger wave	0.084	0.002	663	0.009	−0.052	789
	(0.224)	(0.052)		(0.159)	(0.032)	
A rival deal that never completes						
withdrawn or	0.223	−0.011	365	0.106	−0.009	454
rumoured deal	(0.249)	(0.051)		(0.131)	(0.029)	

Notes: The main design with a deal elsewhere around the firm in place of a designated rival's consolidation: the event is a deal in the channel after a year without one, on firm-years in which no designated rival consolidates, and the comparison firm-years are quiet in the channel for five years and through the horizon. Supplier or customer, partner: a consolidation among the firm's suppliers and customers, or among its partners; rivals of rivals: among the designated rivals of its rivals; industry merger wave: the onset of a year in which the firm's four-digit industry is in the top quarter of its own years of worldwide deal activity. The second group keeps the events followed by no designated rival's consolidation through $t+5$ (a condition on the future, reported as a check). Withdrawn or rumoured deal: a control deal of a designated rival announced or rumoured and never completed, on firm-years without a completed designated rival deal. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

B. News rather than consolidation

If firms responded to the news that a rival seeks a deal rather than to the consolidation itself, a deal announced or rumoured and never completed would produce the same response. Taken as the event, on firm-years without a completed designated rival deal, such deals are followed by +0.22 for the log and +0.11 for the inverse hyperbolic sine, on 365 events and with wide confidence intervals (Table 7). Dating the consolidations at their completion rather than their announcement gives $+0.45^{***}$. Firms respond to consolidations that take place.

C. Innovating to be acquired

When acquirers are active, a small firm may raise its research spending to become a more attractive target (Phillips and Zhdanov 2013). If this motive drove the response, the exposed firms would be acquired more often afterwards. They are not (Table 8). Within five years of the onset, an exposed firm is +1.7 percentage points more likely than a comparison firm to become the target of a control-changing deal, against a comparison rate of 7.5 per cent; among the smallest third of firms, where the motive should be strongest, it is +4.7 points more likely. Without the firms acquired in those five years, the response of log R&D is $+0.47^{***}$, against $+0.49^{***}$ on every event.

D. Rivals named in anticipation

Assumption 5 fails if a firm names competitors because it expects them to consolidate, and the same foresight leads it to invest. The timing of naming speaks against this, and two other checks neither confirm nor rule it out. First, the rival set does not grow ahead of the deal. With the set as the outcome of the main design, the exposed firms' sets were, relative to the year before the onset, larger three and four years earlier than the comparison firms' ($+0.20^{***}$ and $+0.41^{***}$ in the log of one plus the number of rivals), and the exposed firms named new rivals at a slowing pace; in the year of the deal they name many more new rivals ($+0.90^{***}$ in the inverse hyperbolic sine of the count), so naming follows the deal rather than preceding it (Appendix Figure H1). Second, adding the change in the rival set at $t-1$ and $t-2$ to the controls leaves the pooled response at +0.51, close to the main estimate, but on this smaller sample neither it nor any yearly estimate is significant (+0.27 in the third year). Third, the age of the tie does not separate the two readings. When every consolidating rival had been named at least three years before its deal, the pooled response is $+0.85^{***}$; when every one had been named within the two years before, $+0.40^{***}$. Year by year the two are of similar size (+0.23 and $+0.21^{**}$ in the third year). The response to the recent ties is significant from the year of the deal ($+0.05^{**}$), as anticipation would predict, and that to the older ties, as large, is imprecise until the fifth year.

Table 8. Do the exposed firms become targets?

	Firm-years with positive R&D	Firm-years reporting R&D
Panel A. The firm becomes the target of a control-changing deal		
All firms, from $t+1$ to $t+5$	0.017	0.022*
	(0.014)	(0.013)
All firms, from t to $t+5$	0.011	0.016
	(0.015)	(0.014)
Comparison mean, $t+1$ to $t+5$	0.075	0.092
Events; observations	1,961; 4,067	2,509; 5,372
The smallest third by assets, from $t+1$ to $t+5$	0.047	0.053*
	(0.030)	(0.029)
The smallest third by assets, from t to $t+5$	0.035	0.043
	(0.030)	(0.030)
Comparison mean, $t+1$ to $t+5$	0.095	0.108
Events; observations	598; 1,294	700; 1,545
Panel B. The R&D response without the firms acquired from $t+1$ to $t+5$		
Main specification, pooled post-period	0.491***	0.232***
	(0.140)	(0.089)
Main specification, pooled pre-period	−0.005	−0.013
	(0.027)	(0.015)
Events	3,588	4,328
Without the acquired firms, pooled post-period	0.473***	0.221**
	(0.142)	(0.090)
Without the acquired firms, pooled pre-period	−0.002	−0.011
	(0.028)	(0.016)
Events	3,328	3,948

Notes: Panel A: the regression the estimator runs at $h = 0$ under the main specification (same firm-years, controls, fixed effects and clustering), with an indicator that the firm is the target of a control-changing deal in the years shown as the outcome; event years to 2017, so that the five years after are observed. The smallest third: firms in the lowest tercile of total assets at $t-1$, events and comparison firm-years alike. Comparison mean: the share of comparison firm-years that become targets. Panel B: the main specification without the events and comparison firm-years of firms acquired from $t+1$ to $t+5$; pooled post-period over $h = 0$ to 5, pre-period over $h = -4$ to -2. The firm-years with positive R&D are the sample of log R&D, those reporting R&D, zeros included, the sample of its inverse hyperbolic sine. Standard errors clustered by the firm's reporting unit in parentheses. *, **, ***: significance at 10, 5 and 1 per cent.

E. Years without a recorded rival

Two variants address the years without a rival on record. The first keeps only the recorded onsets, those with a rival on record in the year before, and drops the events whose rival set appears in the data with the deal. If the effect came from rival sets recorded because the firm had itself become visible, removing them would shrink it. It grows instead, to $+0.65^{***}$ on 2,102 events, as expected if some of the dropped events were deals inside an episode whose onset went unrecorded. The second variant also keeps only the comparison firm-years with a rival on record at t and $t-1$, so that both sides are measured in the two years that define the event. The response is $+0.35^{**}$ for the log and $+0.28^{***}$ for the inverse hyperbolic sine. The log estimate is smaller than with the full comparison group, by about one standard error, on a narrower group of comparison firms; the estimate for the inverse hyperbolic sine does not move. Within the two years that define the event, the unrecorded years do not produce the effect.

F. Chance and correlated shocks

Firms that name the same rival are exposed to the same deal, and firms in the same industry share shocks that the industry-by-year effects absorb only on average, so the main standard errors could understate the uncertainty. Clustering by four-digit industry, or by the community of the rival network to which the firm belongs, leaves the response of log R&D significant at five per cent or better. Clustering by two-digit industry, with few clusters, gives $p = 0.03$ for the log and $p = 0.13$ for the inverse hyperbolic sine (Table 9).

Randomization inference relies on no cluster structure. Each draw replaces every firm's rival set, year by year, with a random set of the same size drawn among the live rivals of other firms in the firm's four-digit industry, and reruns the main specification. A drawn set has the industry and the size of the real one but no relation of rivalry to the firm. Over 500 draws, the pooled response averages -0.15 for the log, with a standard deviation of 0.21, and -0.07 for the inverse hyperbolic sine (Figure 12). The estimate with the real rival sets lies in the upper tail of both distributions: 0.4 per cent of the draws reach it for the log and 2.0 per cent for the inverse hyperbolic sine. The draws centre below zero, so deals among random firms of the industry are, if anything, followed by lower research spending, the sign that some industry-based studies of rivals find. Counted from zero rather than from the centre of the draws, the two-sided randomization p-values are 0.06 for the log and 0.13 for the inverse hyperbolic sine.

G. The definition of a consolidation, the comparison and the scale

A consolidation includes the purchase of a division or of a set of assets as well as that of a whole company. Keeping only the events whose deals include the acquisition of a whole company gives $+0.44^{***}$; keeping those whose deals are all of this kind, $+0.48^{**}$ on 723

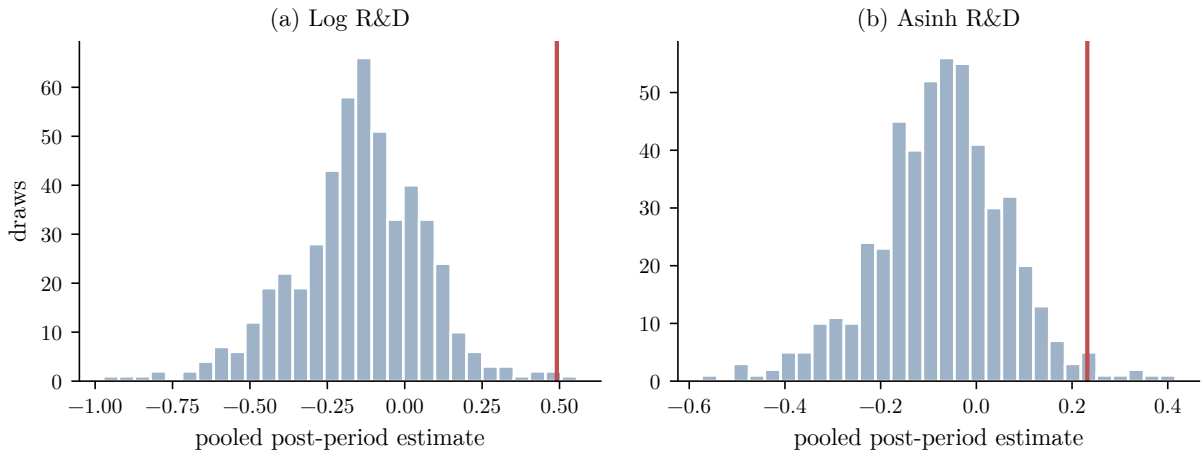


Figure 12. Randomization inference

Notes: Each draw replaces every firm's rival set, year by year, by a random set of the same size drawn among the live rivals of other firms in the firm's four-digit industry, and reruns the main specification with the drawn set: the event is a consolidation in the drawn set after a year without one, the comparison firm-years those whose drawn set has been quiet for five years. 500 draws per outcome; the histogram is the distribution of the pooled post-period estimates and the vertical line the estimate with the real rival sets.

Table 9. Inference on the main estimate

	Log R&D		Asinh R&D	
	Standard error	p-value	Standard error	p-value
Standard errors clustered by				
the firm's reporting unit (main)	0.140	0.000	0.089	0.009
two-digit industry	0.221	0.026	0.155	0.134
four-digit industry	0.157	0.002	0.103	0.024
community of the rival network	0.095	0.000	0.071	0.001
Randomization inference				
share of draws at or above the estimate		0.004		0.020
share of draws as far from zero		0.060		0.132
mean (s.d.) of the draws		−0.147 (0.213)		−0.072 (0.139)

Notes: The pooled post-period estimate of the main specification (+0.491 for log R&D, +0.232 for its inverse hyperbolic sine). Standard errors and two-sided p-values under four clusterings: the firm's reporting unit, as in the main specification; two- and four-digit SIC industry; the community of the rival network to which the firm belongs. Randomization: the share of the drawn estimates (Figure 12) at or above the real one, and at least as far from zero in absolute value; one is added to the numerator and the denominator.

events. The comparison group matters little. With five placebo years instead of four, the response is $+0.49^{***}$; with seven quiet years for the comparison firm-years instead of five, $+0.47^{***}$; against the firms whose rivals have not yet consolidated, $+0.49^{***}$; and against the estimator's standard comparison, which requires the rivals to be quiet only in $t-1$ and t, $+0.52^{***}$ (Appendix Figures D1 and D2).

Finally, the extended two-way fixed-effects estimator of Wooldridge (2023) compares research spending in levels, with a Poisson mean that needs no transformation of the zeros, and takes the firm's first exposure as the event. With the firm's characteristics, it gives a response of $+0.50^{**}$ log points, of the same size as the main estimate. Its pre-exposure years pass a joint test against the not-yet- and never-exposed firms ($p = 0.11$) and are at the margin of rejection against the never-exposed firms alone ($p = 0.05$; Appendix I).

VII. Conclusion

Firms raise their research spending when a rival they name as a competitor enters a control-changing deal. The rise is small at first, about 4 per cent in the year of the deal, and grows to about 24 per cent by the third year, as the firm's assets and workforce expand with it; the data do not pin down how large it becomes later. The response follows the direction of attention: firms respond to the consolidations of the rivals they watch, not to those of the firms that watch them, and most clearly to those of much larger rivals. It does not grow with any of the features through which a merger changes the structure of a market, and deals among the firm's industry peers are followed by no rise in the first three years.

The definition of a firm's rivals is therefore not a detail of measurement. A definition by industry or by market treats rivalry as symmetric and common to every firm in the market; the competitors firms name show that it is neither. Because a firm's response depends on whether it watches the consolidating firm, an estimate built on industry membership averages the response of the firms that watch it with the absence of response of those that do not, and its size depends on the mix. In our data, the early response disappears in the mix. Studies of the firms outside a merger can therefore reach different answers about the same kind of deal when attention is spread differently across their markets, and the same applies to measures of competitive pressure that are common to every firm in an industry.

For merger review, which weighs a deal's effect on innovation (U.S. Department of Justice and Federal Trade Commission 2023), this effect reaches beyond the merging firms and beyond the market in which the deal is assessed. The firms that respond are those that watch the merging firms, and these do not need to share their market: 43 per cent of the named pairs join firms of different two-digit industries, and the response to a consolidating rival outside the firm's two-digit industry, $+0.59^{***}$, is no smaller than the response to one inside its four-digit industry, $+0.47^{***}$. Whether the outsiders' response makes a deal

better or worse for innovation depends on what the extra spending produces. Within five years, it does not raise the number of patents the firms file but widens the range of technologies their patents cover; our estimates measure the response in research spending, not its value.

The evidence does not say why firms respond to the rivals they watch. Keeping up with a rival that has become stronger, learning from its deal about where the market is going, and searching harder when falling behind a peer all predict a response that follows attention, and withdrawn deals show only that the news of a deal is not enough. What the evidence does show is that the pressure a consolidation puts on research spending reaches firms through the rivals they watch, and that the competitors firms name, read in the direction in which they are named, measure that pressure.

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Appendix A. Variables

Table A1 defines the outcomes, the event, the controls of the main specification and the firm groups of Section V. The groups of Section IV, formed by features of the deal and of the consolidating firm, are defined in the text and in the notes to their exhibits.

Table A1. Variable definitions

Variable	Definition
Panel A. Outcomes	
R&D	Research and development expense net of acquired in-process R&D: Compustat's R&D expense less its in-process R&D charge, or FactSet's R&D expense, which excludes the charge (Section II.C). A firm-year without a reported value is missing, not zero.
Log R&D	ln(R&D), on the firm-years with positive R&D.
Asinh R&D	$\text{asinh}(\text{R\&D}) = \ln\left(\text{R\&D} + \sqrt{\text{R\&D}^2 + 1}\right)$, on the firm-years that report R&D, zeros included.
R&D over lagged assets	R&D over the firm's total assets of the previous fiscal year, winsorised.
Patent families originated	The distinct families of utility patents, of any office, with a priority date in the year and an assignee name validated for the firm in that year, without the families that reached the firm through an acquisition; in its inverse hyperbolic sine. Missing for the firms without a validated name, like every patent outcome.
Patents granted	The firm's utility patent applications filed in the year that have been granted; in its inverse hyperbolic sine.
Technology subclasses	The distinct CPC subclasses of the firm's patent families with a priority date in the year; in its inverse hyperbolic sine.
Breadth net of volume	The inverse hyperbolic sine of the links between the patent families of the year and their CPC subclasses, each family counted once in each of its subclasses, less that of the patent families of the year. The patent outcomes, counted by calendar year, enter the fiscal year that contains the calendar year's end.
Panel B. Rivals, deals and the event	
Designated rival	A company the firm names as a competitor in FactSet's records, in the calendar years in which the record is live (Section II.A).
Coverage	The fiscal years from the firm's first record as a source firm in FactSet's relationship data, of any type.
Consolidation	A completed deal that transfers control of a company, a division or a set of assets, with the rival as the buyer or the company bought, dated at its announcement (Section II.B). It reaches the firm in the fiscal year that contains the announcement date.
Rival deal	1 if at least one designated rival enters a consolidation in the fiscal year, 0 otherwise; missing before the firm's coverage begins. A covered year without a live designated rival counts as a year without a rival deal.
Onset	A rival deal in year t after a year $t-1$ without one: the event D_{it} of equation (3).
Comparison firm-year	A firm-year without a rival deal in year t or in any of the five years before it; at horizon h, also without one from $t+1$ to $t+h$.
Share of rivals consolidating	The designated rivals that enter a consolidation in the year over the designated rivals live in the year.
Panel C. Controls, at $t-1$ unless stated	
Lagged change in the outcome	$y_{i,t-1} - y_{i,t-2}$.
Log total assets, log sales	The logarithms of total assets and of sales.

Table A1, continued

Variable	Definition
Log age	$\ln(1 + \text{age})$, with age the fiscal year less the year in which the firm was founded, from FactSet.
Cash over assets	Cash and short-term investments over total assets, winsorised.
Capital expenditure over assets	Capital expenditure over total assets, winsorised.
Asset and sales growth	The change in log total assets and in log sales from $t-2$ to $t-1$.
Live rivals	The number of designated rivals live on any day of the fiscal year.
No rival on record	1 in a covered fiscal year in which no designated rival is live; at t and at $t-1$.
Own consolidation	1 if the firm itself is the buyer or the company bought in a consolidation in the fiscal year; at t, $t-1$ and $t-2$.
Panel D. Firm groups (Section V), before the onset	
R&D stock over assets	$\sum_{k=1}^{5} 0.85^{\,k-1}\,\text{R\&D}_{t-k}$ over total assets at $t-1$. A year without reported R&D counts as zero, and the stock is missing when none of the five years is reported.
Intangible assets over assets	Intangible assets, floored at zero, over total assets, at $t-1$.
Patent stock	The patent families the firm originated from $t-5$ to $t-1$, on the firms matched to patent assignees: none, or a positive stock below or above its median.
Total assets, cash over assets	At $t-1$, as in Panel C.
Leverage	Long-term and short-term debt over total assets, floored at zero and capped at the 99th percentile of the fiscal year, at $t-1$.
Stock of rival consolidation	The sum, from $t-5$ to $t-1$, of the share of rivals consolidating, on the events whose rival deals are observed in each of these years; $t-1$ is quiet for every event. None, or a positive stock below or above its median.
Years since the last rival deal	Two, three or four, or five or more (including none in the five years), on the same events.

Notes: Amounts are in millions of US dollars, converted at the rate of the period's end for balance-sheet items and at the period's average rate for flows. Winsorised: at the 1st and 99th percentiles of the fiscal year, over the firm-years of the window with the firm's own accounts. The firm groups of Panel D are cut into terciles, or at the median, over the estimation firm-years.

Appendix B. The exposure mappings

Table B1 lists the exposure mappings the paper estimates (Section I.A). Each row changes one part of the main mapping and keeps the rest.

Table B1. The exposure mappings of the paper

	Section
The main mapping	
The designated rivals; every consolidation; the onset of an episode; five quiet years before t and none through $t + h$	III
The reference set changes	
Rivals named one way; mutual rivals; firms that name the firm without being named back, each with the other two sets' consolidations in the controls	IV.A
Four-digit industry peers	IV.E
Suppliers and customers; partners; rivals of rivals; the firm's four-digit industry in a merger wave	VI.A
Random sets of the same size, drawn among the live rivals of other firms in the firm's four-digit industry	VI.F
The exposed state is split by the deals	
The size of the consolidating firm relative to the firm, in each direction of naming	IV.B
The age of the tie	IV.C, VI.D
The product similarity of the two firms	IV.D
The share of the rival set that consolidates; the size of the deal relative to the acquiring rival; one rival acquiring another; the rival as buyer or company bought; industry proximity; distance and country; merger waves	IV.E
Acquisitions of a whole company	VI.G
The deals that count, their dating and the quiet state change	
Deals announced and later withdrawn, or rumoured, in place of completed ones; consolidations dated at completion	VI.B
Onsets with a rival on record in the year before	VI.E
Seven quiet years; firms not yet exposed; the estimator's standard comparison; the first exposure as an absorbing event	VI.G
Other mappings' exposures in the controls	
The consolidations of suppliers and customers, partners and rivals of rivals; the industry's merger wave	VI.A

Notes: The main mapping is given by its reference set, the deals that count, the event and the comparison (Section I.A). Each other row changes one part of it and keeps the rest of the main specification, except the first exposure as an absorbing event, which is estimated with the extended two-way fixed-effects estimator of Wooldridge (2023). Each part of a split exposed state is compared with the quiet state of the mapping it splits, so that the estimates differ only in the events they contain. The groups of Section V split the events of the main mapping by the characteristics of the firm rather than by the deals, and are not listed.

Appendix C. Data

Table C1 sets the named pairs against the text-based network industries of Hoberg and Phillips (2016), in their embeddings-based (ETNIC-3) and word-based (TNIC-3) versions. Panel A gives the share of the observable named pairs that the network relates, with the pairs that share an industry code and random pairs as benchmarks; Panel B shows how few of a firm's network relationships it names; Panel C shows how many named pairs lie outside the network's universe of US-listed firms.

Table C1. The named rivals against Hoberg and Phillips's text-based network

	Embeddings-based (ETNIC-3)	Word-based (TNIC-3)
Panel A. Agreement		
Named pairs related in the network	71	60
named one way	65	55
mutual	91	78
Pairs in the same four-digit SIC industry	52	51
Pairs in the same three-digit SIC industry	41	41
Random pairs of our firms	3	3
Panel B. Selection		
Network rivals per firm-year (median, number)	101	41
of which named (median, number)	3	2
Network relationships that are named	2	2
Similarity above threshold, named pairs (median)	0.168	0.069
Similarity above threshold, other pairs (median)	0.072	0.056
Panel C. Coverage of the named pairs		
Focal firm outside the United States	42	
Rival outside the United States	49	
Rival not publicly listed	42	
Named in one direction only	89	
Named pairs observable in the network	22	21
Event pairs observable in the network	33	32

Notes: ETNIC-3: the embeddings-based version of the network, the latest its authors distribute; TNIC-3: the word-based version used in most published work (Hoberg and Phillips 2016). Designated rival pair-years 2003–2021. Panel A: among the pairs whose two firms are both in the network's universe in the year, the share related in the network; for comparison, the same share for pairs of our firms that share a four- or three-digit SIC code, and for random pairs of our firms. Panel B: the text-based relationships of our firms, and how many of them the firms name; similarity in excess of the network's threshold. Panel C: what the network cannot see. Per cent unless stated.

Appendix D. The comparison group and the placebo years

The main comparison firm-years are quiet for five years before t, a condition the estimator does not impose. Figure D1 drops it, and the pooled response of log R&D is $+0.52^{***}$. Figure D2 strengthens it. With comparison firm-years quiet for seven years, the pooled response is $+0.47^{***}$ for the log and $+0.24^{***}$ for the inverse hyperbolic sine; against the firms not yet exposed to a rival's consolidation, it is $+0.49^{***}$ and $+0.24^{***}$. The pooled placebo estimates of the log stay at zero in both, -0.000 and $+0.002$.

Table D1 extends the placebo years back to $h = -5$. The estimate at $h = -5$ is -0.07 for the log and -0.06 for the inverse hyperbolic sine, and the pooled placebo estimate over $h = -5$ to -2 is -0.025 and -0.024. Five years before the onset, the exposed and comparison firms were on the same path.

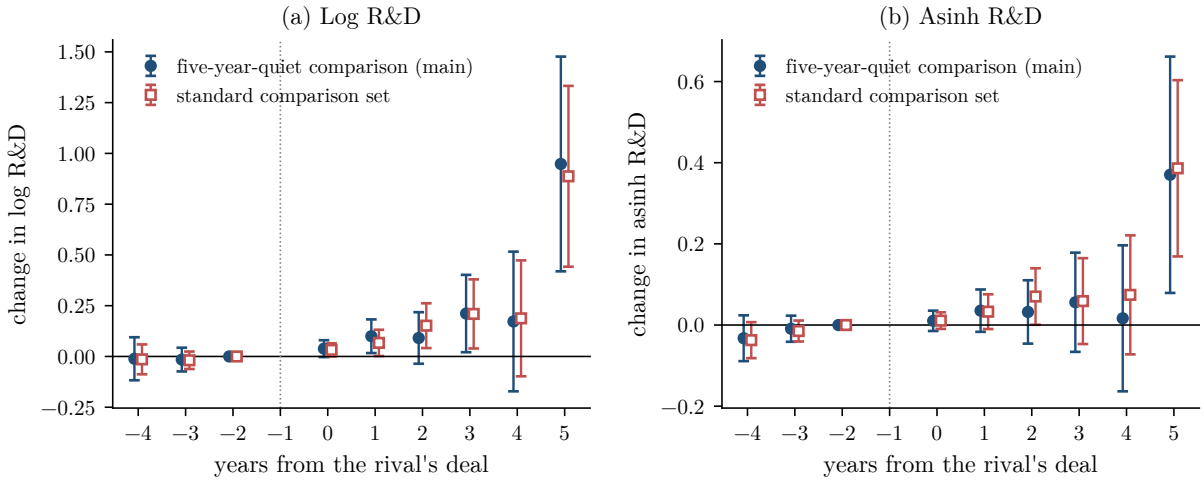


Figure D1. The estimator's standard comparison set

Notes: The main specification, and the same against the comparison firm-years the estimator keeps by construction once the event is the onset: no rival deal at $t-1$, at t and through the horizon, without the five quiet years before t. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

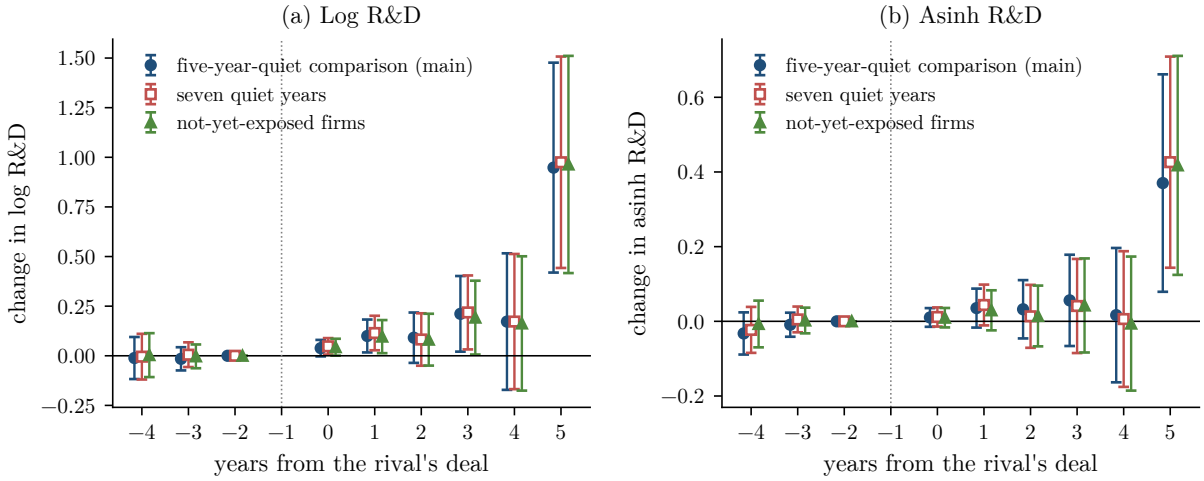


Figure D2. Other comparison groups

Notes: The main specification, and the same against comparison firm-years whose rivals made no deal in the seven years before t rather than five (seven quiet years), and against the firm-years of firms that have not yet been exposed to a rival's consolidation, quiet for five years as in the main comparison (the estimator's not-yet-treated option). The events are unchanged. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

Table D1. A fifth placebo year

	Log R&D		Inverse hyperbolic sine of R&D	
Horizon h	Estimate	Observations	Estimate	Observations
-5	-0.073	1,922	-0.064	2,588
	(0.090)		(0.049)	
-4	-0.011	2,660	-0.033	3,470
	(0.054)		(0.029)	
-3	-0.015	4,114	-0.009	5,201
	(0.030)		(0.016)	
-2	0.000	6,954	0.000	8,461
0	0.038*	7,015	0.010	8,554
	(0.021)		(0.013)	
1	0.100**	4,546	0.035	5,621
	(0.042)		(0.027)	
2	0.091	2,767	0.032	3,496
	(0.065)		(0.040)	
3	0.211**	1,819	0.056	2,332
	(0.097)		(0.062)	
4	0.172	1,178	0.017	1,529
	(0.175)		(0.092)	
5	0.948***	785	0.370**	1,023
	(0.270)		(0.149)	
Pooled, -5 to -2	-0.025	1,885	-0.024	2,555
	(0.046)		(0.026)	
Pooled, 0 to 5	0.491***	756	0.232***	999
	(0.140)		(0.089)	
Events at $h = 0$	3,588		4,328	

Notes: The main specification with the placebo horizons extended to $h = -5$; the other horizons are those of the main estimates. Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; Observations: those of that horizon's regression, events and comparison firm-years together. The estimate at $h = -2$ is zero by construction, because the lagged change in the outcome is a control. The pooled post-period row averages $h = 0$ to 5 on the firm-years observed through $h = 5$; the pooled pre-period row averages $h = -5$ to -2 on the firm-years observed in each of them. *, **, ***: significance at 10, 5 and 1 per cent.

Appendix E. The main estimates

Table E1 gives the estimates of Figures 3 and 5 at every horizon, with the observations of each regression. Table E2 takes apart the rival-set block of Table 3, adding one of its parts at a time to column (3). Figure E1 estimates the response on the firm-years observed at every horizon, and Figure E2 on the comparison firm-years of the countries, industries and sizes that hold an event in the same year.

Tables E3 and E4 report the main estimates under three changes, one at a time: without the country-by-year effects; with region-by-year effects in their place, for the regions by which FactSet dates its coverage; and with the comparison firm-years without a rival on record at t dropped rather than given an indicator. The estimates of the first two years barely move; those of the fourth and fifth years, which rest on few firm-years spread over many fixed-effect cells, move by as much as their standard errors. The pooled estimate over $h = 0$ to 3, on the firm-years observed through the third year, is +0.05 for the log and +0.01 for the inverse hyperbolic sine. This average is small, although the third-year estimate, on nearly the same firm-years, is $+0.21^{**}$: like the six-year pooled estimate, it describes the firm-years observed throughout and weighs each year equally.

Table E1. The main estimates by horizon

	Log R&D		Inverse hyperbolic sine of R&D		R&D over lagged assets	
Horizon h	Estimate	Observations	Estimate	Observations	Estimate	Observations
-4	-0.011	2,660	-0.033	3,470	0.007	2,750
	(0.054)		(0.029)		(0.006)	
-3	-0.015	4,114	-0.009	5,201	0.005	3,766
	(0.030)		(0.016)		(0.004)	
-2	0.000	6,954	0.000	8,461	0.000	5,626
0	0.038*	7,015	0.010	8,554	0.002	5,768
	(0.021)		(0.013)		(0.002)	
1	0.100**	4,546	0.035	5,621	0.004	3,864
	(0.042)		(0.027)		(0.005)	
2	0.091	2,767	0.032	3,496	-0.001	2,499
	(0.065)		(0.040)		(0.012)	
3	0.211**	1,819	0.056	2,332	0.003	1,709
	(0.097)		(0.062)		(0.018)	
4	0.172	1,178	0.017	1,529	-0.017	1,149
	(0.175)		(0.092)		(0.021)	
5	0.948***	785	0.370**	1,023	0.051***	771
	(0.270)		(0.149)		(0.020)	
Pooled, -4 to -2	-0.005	2,638	-0.013	3,450	0.004	2,735
	(0.027)		(0.015)		(0.004)	
Pooled, -4 to -3	-0.008	2,638	-0.019	3,450	0.007	2,735
	(0.040)		(0.023)		(0.005)	
Pooled, 0 to 5	0.491***	756	0.232***	999	0.036**	759
	(0.140)		(0.089)		(0.017)	
Events at $h = 0$	3,588		4,328		3,138	

Notes: Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; Observations: those of that horizon's regression, events and comparison firm-years together. The estimate at $h = -2$ is zero by construction, because the lagged change in the outcome is a control. The pooled post-period row averages $h = 0$ to 5 on the firm-years observed through $h = 5$; the pooled pre-period rows average the placebo years, the second over the informative ones only. *, **, ***: significance at 10, 5 and 1 per cent.

Table E2. The rival-set block taken apart

	Log R&D			Inverse hyperbolic sine of R&D		
	After	Before	Events	After	Before	Events
Firm and growth controls, column (3)	0.110	−0.019	3,588	0.085*	−0.032***	4,328
	(0.079)	(0.020)		(0.049)	(0.012)	
Adding one part of the rival-set block						
the size of the live set at $t-1$	0.183*	−0.021	3,588	0.137**	−0.023*	4,328
	(0.095)	(0.024)		(0.057)	(0.014)	
both no-record indicators	0.476***	0.002	3,588	0.205**	−0.015	4,328
	(0.138)	(0.025)		(0.084)	(0.014)	
the no-record indicator at t	0.516***	−0.001	3,588	0.227***	−0.016	4,328
	(0.144)	(0.024)		(0.079)	(0.014)	
the no-record indicator at $t-1$	0.161*	−0.016	3,588	0.110**	−0.028**	4,328
	(0.083)	(0.021)		(0.049)	(0.012)	
The whole block, column (4)	0.493***	−0.004	3,588	0.234***	−0.011	4,328
	(0.143)	(0.027)		(0.090)	(0.016)	

Notes: Column (3) of Table 3 (two-digit-industry-by-year and country-by-year fixed effects, the lagged change in the outcome, the firm's characteristics and growth) with one part of the rival-set block added at a time: the number of live designated rivals at $t-1$; the indicators for a year without a rival on record at t and at $t-1$; the indicator at t alone, which varies only among comparison firm-years, since an event requires a recorded rival deal; the indicator at $t-1$ alone. The last row is column (4), the whole block. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

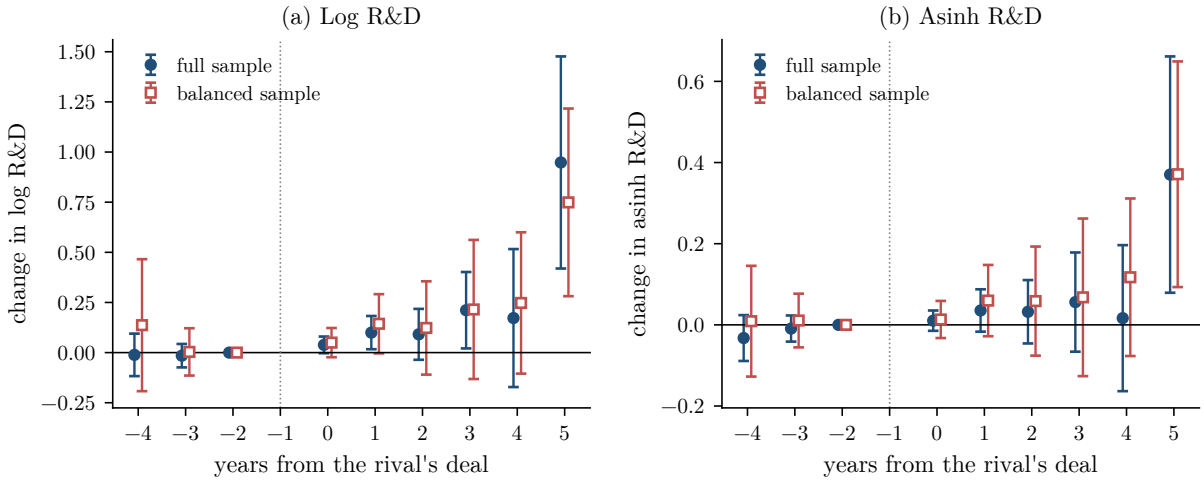


Figure E1. The balanced sample

Notes: The main specification, and the same on the firm-years whose outcome is observed in every year from $t+1$ to $t+5$, events and comparison firm-years alike, so that every horizon rests on the same firms. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

Table E3. The response of log R&D under other fixed effects and comparison groups

	Main	Industry-year effects only	Industry-year and region-year effects	No-record comparisons dropped
−4	−0.011	−0.027	0.001	0.002
	(0.054)	(0.045)	(0.051)	(0.062)
−3	−0.015	−0.023	−0.020	0.022
	(0.030)	(0.027)	(0.028)	(0.035)
−2	0.000	0.000	0.000	0.000
0	0.038*	0.031	0.035*	0.031
	(0.021)	(0.020)	(0.020)	(0.024)
1	0.100**	0.070*	0.101**	0.086
	(0.042)	(0.040)	(0.043)	(0.052)
2	0.091	0.088	0.063	0.018
	(0.065)	(0.059)	(0.066)	(0.087)
3	0.211**	0.177**	0.122	0.113
	(0.097)	(0.083)	(0.098)	(0.162)
4	0.172	0.128	−0.092	0.095
	(0.175)	(0.140)	(0.171)	(0.275)
5	0.948***	0.453**	0.447*	1.450***
	(0.270)	(0.191)	(0.230)	(0.411)
Pooled, −4 to −2	−0.005	−0.014	−0.005	−0.000
	(0.027)	(0.023)	(0.025)	(0.032)
Pooled, 0 to 5	0.491***	0.285***	0.294**	0.472*
	(0.140)	(0.101)	(0.119)	(0.253)
Events at $h = 0$	3,588	3,588	3,588	3,588

Notes: Main: the main specification (Table E1), with two-digit-industry-by-year and country-by-year fixed effects. Industry-year effects only: the country-by-year effects dropped. Industry-year and region-year effects: the country-by-year effects replaced by effects for each of FactSet's coverage regions in each year. No-record comparisons dropped: the comparison firm-years without a rival on record at t removed, instead of given an indicator, which then leaves the controls. Each column changes one element of the main specification. Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; the estimate at $h = -2$ is zero by construction. The pooled rows average the placebo years and $h = 0$ to 5, the latter on the firm-years observed through $h = 5$. *, **, ***: significance at 10, 5 and 1 per cent.

Table E4. The response of the inverse hyperbolic sine of R&D under other fixed effects and comparison groups

	Main	Industry-year effects only	Industry-year and region-year effects	No-record comparisons dropped
-4	-0.033	-0.016	-0.034	-0.039
	(0.029)	(0.027)	(0.028)	(0.033)
-3	-0.009	-0.002	-0.015	0.010
	(0.016)	(0.015)	(0.015)	(0.019)
-2	0.000	0.000	0.000	0.000
0	0.010	0.006	0.008	-0.003
	(0.013)	(0.012)	(0.012)	(0.014)
1	0.035	0.019	0.042	0.043
	(0.027)	(0.025)	(0.026)	(0.031)
2	0.032	0.012	0.016	0.012
	(0.040)	(0.037)	(0.043)	(0.043)
3	0.056	0.063	0.042	0.021
	(0.062)	(0.054)	(0.067)	(0.066)
4	0.017	0.097	-0.028	-0.062
	(0.092)	(0.081)	(0.093)	(0.122)
5	0.370**	0.252**	0.215*	0.537**
	(0.149)	(0.112)	(0.127)	(0.229)
Pooled, -4 to -2	-0.013	-0.001	-0.011	-0.014
	(0.015)	(0.014)	(0.015)	(0.018)
Pooled, 0 to 5	0.232***	0.157**	0.142*	0.359***
	(0.089)	(0.070)	(0.078)	(0.128)
Events at $h = 0$	4,328	4,328	4,328	4,328

Notes: Main: the main specification (Table E1), with two-digit-industry-by-year and country-by-year fixed effects. Industry-year effects only: the country-by-year effects dropped. Industry-year and region-year effects: the country-by-year effects replaced by effects for each of FactSet's coverage regions in each year. No-record comparisons dropped: the comparison firm-years without a rival on record at t removed, instead of given an indicator, which then leaves the controls. Each column changes one element of the main specification. Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; the estimate at $h = -2$ is zero by construction. The pooled rows average the placebo years and $h = 0$ to 5, the latter on the firm-years observed through $h = 5$. *, **, ***: significance at 10, 5 and 1 per cent.

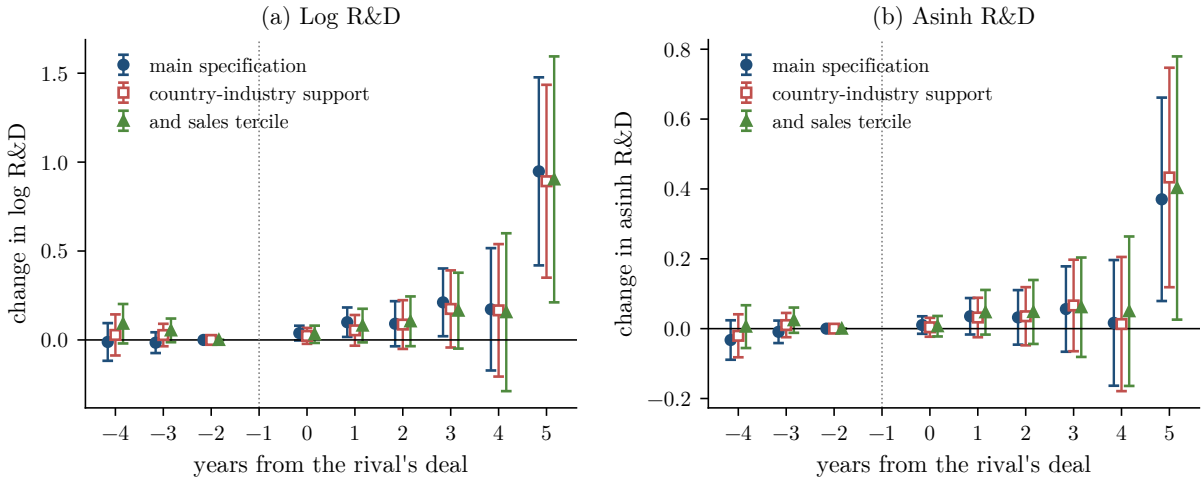


Figure E2. Common support

Notes: The main specification, and the same with the comparison firm-years restricted to the cells that hold at least one event in the same year: the same country and two-digit industry (country-industry support), and in addition the same tercile of sales. The events are unchanged. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

Appendix F. Patents

Tables F1 and F2 give the estimates of Figure 6 at every horizon. Figure F1 extends the four patent outcomes to eight years after the onset. The patent families the firm originates, negative through the sixth year (−0.26), turn positive in the seventh and the eighth (+0.19 and +0.80); granted patents rise to +0.51 and +0.55, and technology subclasses to +1.01** in the eighth year. Pooled over the nine years from the onset, granted patents are +0.63** higher and subclasses +0.63**. Breadth net of volume, which rose to the fourth year, does not differ from zero afterwards: it is −0.14, +0.14 and +0.15 in the sixth to the eighth year.

These horizons rest on few firm-years: the regression at $h = 8$ has 392 observations, against 9,874 at the onset. An event reaches the eighth year only if it is old enough and no new onset intervenes, and a comparison firm-year only if its rivals make no deal from $t - 5$ to $t + 8$, so the later horizons rest on different firms from the earlier ones. Tables F3 and F4 show how much this matters. They compare, horizon by horizon, the full sample with two samples that follow the same firms for seven years: the events and comparison firm-years whose patent families are observed in every year to $t + 7$, and, among these, the fixed cohort of events without a new onset and comparison firm-years without a rival deal through $t + 7$, in which every horizon rests on the same firm-years. In the third year, the patent families the firm originates are −0.17** in the full sample, −0.07 among the firms observed for seven years and +0.60** in the fixed cohort; granted patents are −0.05, −0.04 and +0.82**. The fixed cohort's estimates are large and positive from the second year on, but its placebo years are imprecise for families (+0.25, pooled) and depart from

Table F1. The patent estimates by horizon

	Patent families originated		Patents granted	
Horizon h	Estimate	Observations	Estimate	Observations
-4	-0.070	4,406	-0.026	4,406
	(0.043)		(0.049)	
-3	0.013	6,249	-0.047	6,249
	(0.032)		(0.032)	
-2	0.000	9,571	0.000	9,571
0	-0.025	9,874	0.010	9,874
	(0.025)		(0.024)	
1	-0.039	6,786	-0.019	6,786
	(0.041)		(0.040)	
2	-0.134^{**}	4,440	-0.054	4,440
	(0.062)		(0.062)	
3	-0.173^{**}	3,064	-0.051	3,064
	(0.082)		(0.094)	
4	-0.117	2,035	-0.133	2,035
	(0.112)		(0.121)	
5	-0.172	1,332	-0.028	1,332
	(0.152)		(0.174)	
Pooled, -4 to -2	-0.021	4,406	-0.033	4,406
	(0.024)		(0.025)	
Pooled, 0 to 5	-0.081	1,320	0.050	1,320
	(0.108)		(0.114)	
Events at $h = 0$	5,004		5,004	

Notes: Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; Observations: those of that horizon's regression, events and comparison firm-years together. The estimate at $h = -2$ is zero by construction, because the lagged change in the outcome is a control. The pooled post-period row averages $h = 0$ to 5 on the firm-years observed through $h = 5$; the pooled pre-period rows average the placebo years, the second over the informative ones only. *, **, ***: significance at 10, 5 and 1 per cent.

zero for granted patents, by -1.56^{**} at $h = -4$ and -0.54^{**} at $h = -3$; the firms observed for seven years show the same departure for granted patents (-0.30^{***} at $h = -4$). The firms that can be followed this long already differed from their comparison firms in their patenting before the onset, and the later horizons mix any lag between research spending and patents with the selection of these firms. We therefore read the rise in patenting at the end of the horizon as consistent with a long lag, not as an estimate of it.

Table F2. The patent estimates by horizon, continued

	Technology subclasses		Breadth net of volume	
Horizon h	Estimate	Observations	Estimate	Observations
-4	-0.049	4,406	-0.007	4,406
	(0.041)		(0.024)	
-3	0.020	6,249	-0.001	6,249
	(0.030)		(0.018)	
-2	0.000	9,571	0.000	9,571
0	-0.000	9,874	0.019	9,874
	(0.025)		(0.014)	
1	0.025	6,786	0.060***	6,786
	(0.040)		(0.021)	
2	0.000	4,440	0.076**	4,440
	(0.063)		(0.032)	
3	-0.039	3,064	0.083*	3,064
	(0.085)		(0.043)	
4	0.035	2,035	0.132**	2,035
	(0.117)		(0.062)	
5	-0.110	1,332	0.053	1,332
	(0.160)		(0.082)	
Pooled, -4 to -2	0.003	4,406	0.013	4,406
	(0.023)		(0.013)	
Pooled, 0 to 5	-0.026	1,320	0.079	1,320
	(0.101)		(0.060)	
Events at $h = 0$	5,004		5,004	

Notes: Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; Observations: those of that horizon's regression, events and comparison firm-years together. The estimate at $h = -2$ is zero by construction, because the lagged change in the outcome is a control. The pooled post-period row averages $h = 0$ to 5 on the firm-years observed through $h = 5$; the pooled pre-period rows average the placebo years, the second over the informative ones only. *, **, ***: significance at 10, 5 and 1 per cent.

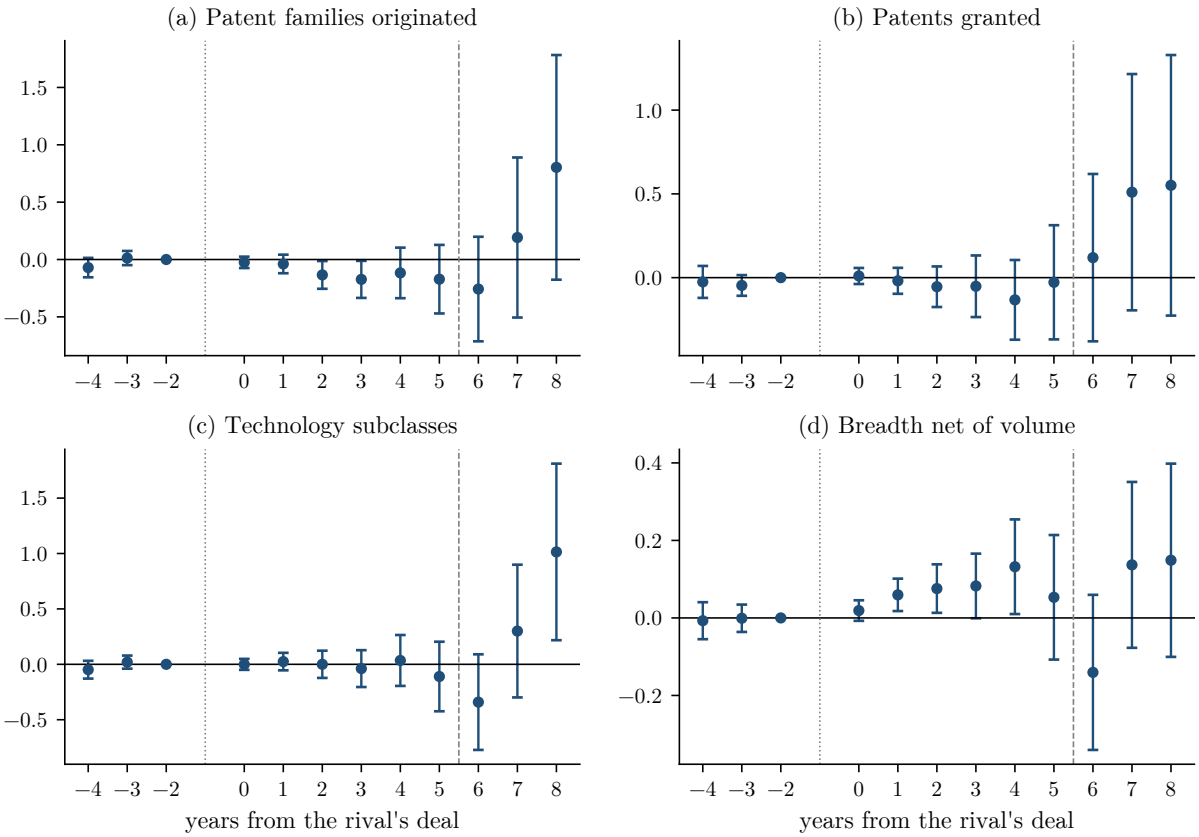


Figure F1. The patent outcomes to eight years after the onset

Notes: The main specification of Figure 6, with the horizons extended to $h = 8$; the dashed line marks the end of the horizons the text reads. 5,004 events at $h = 0$; the regressions rest on 9,874 observations at $h = 0$ and on 913, 603 and 392 at $h = 6$, 7 and 8. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

Table F3. The seven-year samples: patent families originated

	Full sample		Observed seven years		Fixed cohort	
Horizon h	Estimate	Observations	Estimate	Observations	Estimate	Observations
-4	-0.070	4,406	0.020	1,209	0.325	309
	(0.043)		(0.096)		(0.699)	
-3	0.013	6,249	0.043	1,602	0.275	415
	(0.032)		(0.066)		(0.232)	
-2	0.000	9,571	0.000	2,411	0.000	572
0	-0.025	9,874	0.006	2,472	0.153	591
	(0.025)		(0.050)		(0.208)	
1	-0.039	6,786	-0.125	1,967	0.086	591
	(0.041)		(0.084)		(0.246)	
2	-0.134^{**}	4,440	-0.200^{*}	1,604	0.525^{*}	591
	(0.062)		(0.119)		(0.278)	
3	-0.173^{**}	3,064	-0.072	1,337	0.595^{**}	591
	(0.082)		(0.129)		(0.274)	
4	-0.117	2,035	-0.173	1,102	0.855^{***}	591
	(0.112)		(0.199)		(0.330)	
5	-0.172	1,332	-0.081	904	0.894^{***}	591
	(0.152)		(0.218)		(0.320)	
6	-0.257	913	-0.176	748	0.887^{**}	591
	(0.233)		(0.255)		(0.383)	
7	0.192	603	0.304	591	0.304	591
	(0.356)		(0.357)		(0.357)	
Pooled, -4 to -2	-0.021	4,406	0.019	1,209	0.250	309
	(0.024)		(0.050)		(0.332)	
Pooled, 0 to 7	0.537^{**}	591	0.537^{**}	591	0.537^{**}	591
	(0.260)		(0.260)		(0.260)	
Events at $h = 0$	5,004		1,245		674	

Notes: Full sample: the main specification on the firms matched to patent assignees, to $h = 7$. Observed seven years: the events and comparison firm-years whose patent families are observed in every year from $t + 1$ to $t + 7$; at each horizon the estimator still drops the comparison firm-years that become exposed and the events followed by a new onset. Fixed cohort: of these, the events without a new onset and the comparison firm-years without a rival deal through $t + 7$, so that every horizon rests on the same firm-years. Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; Observations: those of that horizon's regression, events and comparison firm-years together. The estimate at $h = -2$ is zero by construction, because the lagged change in the outcome is a control. The pooled post-period row averages $h = 0$ to 7 on the firm-years observed and clean through $h = 7$, which are those of the fixed cohort, so that it is the same in the three columns; the pooled pre-period row averages the placebo years. *, **, ***: significance at 10, 5 and 1 per cent.

Table F4. The seven-year samples: patents granted

	Full sample		Observed seven years		Fixed cohort	
Horizon h	Estimate	Observations	Estimate	Observations	Estimate	Observations
-4	-0.026	4,406	-0.301^{***}	1,209	-1.559^{**}	309
	(0.049)		(0.111)		(0.662)	
-3	-0.047	6,249	-0.166^{**}	1,602	-0.542^{**}	415
	(0.032)		(0.078)		(0.261)	
-2	0.000	9,571	0.000	2,411	0.000	572
0	0.010	9,874	-0.052	2,472	0.375^{*}	591
	(0.024)		(0.054)		(0.192)	
1	-0.019	6,786	-0.023	1,967	0.733^{**}	591
	(0.040)		(0.091)		(0.288)	
2	-0.054	4,440	-0.096	1,604	0.428	591
	(0.062)		(0.136)		(0.339)	
3	-0.051	3,064	-0.039	1,337	0.818^{**}	591
	(0.094)		(0.138)		(0.335)	
4	-0.133	2,035	-0.048	1,102	1.060^{***}	591
	(0.121)		(0.179)		(0.316)	
5	-0.028	1,332	0.036	904	1.370^{***}	591
	(0.174)		(0.226)		(0.333)	
6	0.119	913	0.132	748	0.974^{**}	591
	(0.255)		(0.293)		(0.408)	
7	0.510	603	0.634^{*}	591	0.634^{*}	591
	(0.360)		(0.358)		(0.358)	
Pooled, -4 to -2	-0.033	4,406	-0.138^{**}	1,209	-0.974^{**}	309
	(0.025)		(0.060)		(0.380)	
Pooled, 0 to 7	0.799^{***}	591	0.799^{***}	591	0.799^{***}	591
	(0.289)		(0.289)		(0.289)	
Events at $h = 0$	5,004		1,245		674	

Notes: Full sample: the main specification on the firms matched to patent assignees, to $h = 7$. Observed seven years: the events and comparison firm-years whose patent families are observed in every year from $t + 1$ to $t + 7$; at each horizon the estimator still drops the comparison firm-years that become exposed and the events followed by a new onset. Fixed cohort: of these, the events without a new onset and the comparison firm-years without a rival deal through $t + 7$, so that every horizon rests on the same firm-years. Estimate: the change in the outcome from the year before the onset to year h for the exposed firm-years relative to the comparison firm-years, with its standard error, clustered by the firm's reporting unit, in parentheses; Observations: those of that horizon's regression, events and comparison firm-years together. The estimate at $h = -2$ is zero by construction, because the lagged change in the outcome is a control. The pooled post-period row averages $h = 0$ to 7 on the firm-years observed and clean through $h = 7$, which are those of the fixed cohort, so that it is the same in the three columns; the pooled pre-period row averages the placebo years. *, **, ***: significance at 10, 5 and 1 per cent.

Appendix G. Whose consolidation and which firms

Tables G1 to G4 give the estimates plotted in Figures 8, 9 and 10: for each group, the pooled post-period and placebo estimates of log R&D, of its inverse hyperbolic sine and of R&D over lagged assets, with the number of events. Tables G2 and G3 split the firm groups of Figure 9 as Sections V.A and V.B do. The text reads no estimate whose placebo years depart from zero.

Table G1. The response by whom the firm watches

	Log R&D			Inverse hyperbolic sine of R&D			R&D over lagged assets		
	After	Before	Events	After	Before	Events	After	Before	Events
Named rivals	0.491***	−0.005	3,588	0.232***	−0.013	4,328	0.036**	0.004	3,138
(main)	(0.140)	(0.027)		(0.089)	(0.015)		(0.017)	(0.004)	
By the direction of naming									
rivals the firm	0.368***	0.004	3,725	0.185***	−0.006	4,501	0.014	0.002	3,338
names, one way	(0.117)	(0.023)		(0.064)	(0.014)		(0.009)	(0.003)	
mutual rivals	0.172***	0.006	3,777	0.097**	0.002	4,276	−0.008	−0.004	4,012
	(0.059)	(0.012)		(0.044)	(0.010)		(0.009)	(0.004)	
firms that name	0.080	0.015	4,166	0.056	0.012	4,808	−0.029***	−0.001	4,646
the firm, one way	(0.059)	(0.011)		(0.047)	(0.009)		(0.009)	(0.003)	
Consolidating firm over ten times the firm's sales									
rival named	0.424***	0.004	2,312	0.225***	−0.009	2,622	0.021*	0.005	1,812
one way	(0.125)	(0.026)		(0.075)	(0.017)		(0.012)	(0.004)	
firm naming the	−0.013	−0.003	430	−0.038	−0.003	476	−0.017	0.004	461
firm one way	(0.073)	(0.030)		(0.059)	(0.022)		(0.013)	(0.008)	
Consolidating firm one to ten times the firm's sales									
rival named	0.157	0.020	851	0.102	0.002	1,140	−0.001	−0.003	887
one way	(0.138)	(0.034)		(0.071)	(0.021)		(0.011)	(0.004)	
mutual rival	0.154**	0.000	1,776	0.072	−0.004	2,031	−0.015	−0.008**	1,888
	(0.067)	(0.015)		(0.051)	(0.011)		(0.010)	(0.004)	
firm naming the	0.119*	0.042***	1,367	0.082	0.033***	1,577	−0.028***	0.002	1,526
firm one way	(0.064)	(0.016)		(0.051)	(0.012)		(0.010)	(0.004)	
Age of the oldest consolidating tie									
named 0–1	0.376***	0.037	2,035	0.181**	0.004	2,363	0.035*	0.007	1,187
years before	(0.126)	(0.038)		(0.084)	(0.022)		(0.018)	(0.006)	
2–4 years	0.833***	−0.062*	765	0.394***	−0.043**	927	0.012	0.007	926
	(0.233)	(0.034)		(0.121)	(0.021)		(0.025)	(0.006)	
5 years or more	0.581**	0.024	707	0.254*	0.007	934	0.052**	0.001	934
	(0.227)	(0.038)		(0.130)	(0.022)		(0.021)	(0.004)	
Product similarity of the consolidating pair									
high	0.468**	0.074	308	0.208*	0.017	504	0.029	0.002	415
	(0.202)	(0.049)		(0.115)	(0.023)		(0.022)	(0.006)	
low	0.472**	−0.027	287	0.270**	−0.028	421	0.023	0.008	335
	(0.206)	(0.050)		(0.117)	(0.026)		(0.024)	(0.007)	
related by	0.672***	−0.009	231	0.337**	−0.047	294	0.085***	0.011	237
naming only	(0.242)	(0.051)		(0.141)	(0.033)		(0.030)	(0.010)	
not observable	0.457***	−0.015	2,762	0.197**	−0.011	3,109	0.030*	0.003	2,151
	(0.120)	(0.028)		(0.084)	(0.017)		(0.017)	(0.003)	

Notes: Direction: three disjoint sets of competitors, each with the main design (Figure 7). Size: the same designs on the events whose largest consolidating firm has more than ten, or between one and ten, times the firm's sales in the year before. Age of the oldest tie: the main specification on the events grouped by how long the firm had named its longest-standing consolidating rival, truncated by the start of its records. Product similarity: the main specification on the events grouped by the largest similarity between the firm and a consolidating rival in the embeddings-based text network, above or below the median of the related pairs, below the network's threshold, or outside its universe. After: the pooled post-period estimate, the average change in the outcome from $h = 0$ to $h = 5$ relative to the year before the deal, on the firm-years observed through $h = 5$; Before: the pooled placebo estimate over the horizons $h = -4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h = 0$. *, **, ***: significance at 10, 5 and 1 per cent.

Table G2. The response by the firm's innovation assets

	Log R&D			Inverse hyperbolic sine of R&D			R&D over lagged assets		
	After	Before	Events	After	Before	Events	After	Before	Events
Named rivals	0.491***	−0.005	3,588	0.232***	−0.013	4,328	0.036**	0.004	3,138
(main)	(0.140)	(0.027)		(0.089)	(0.015)		(0.017)	(0.004)	
R&D stock over total assets									
lowest third	0.344**	0.028	1,280	0.136	0.008	2,009	0.042**	0.002	1,410
	(0.165)	(0.051)		(0.087)	(0.020)		(0.017)	(0.004)	
middle third	0.515***	−0.003	1,697	0.314***	−0.018	1,705	0.036**	0.006	1,203
	(0.142)	(0.031)		(0.096)	(0.020)		(0.017)	(0.004)	
highest third	0.544***	−0.036	611	0.217	−0.038	614	0.030	0.006	525
	(0.206)	(0.037)		(0.133)	(0.027)		(0.030)	(0.009)	
Intangible assets over total assets									
lowest third	0.645***	−0.009	587	0.290**	−0.005	773	0.034	0.016	608
	(0.201)	(0.042)		(0.117)	(0.025)		(0.027)	(0.009)	
middle third	0.503***	0.011	1,303	0.289***	−0.012	1,500	0.050***	0.004	1,066
	(0.135)	(0.036)		(0.090)	(0.020)		(0.018)	(0.004)	
highest third	0.399**	−0.014	1,655	0.168	−0.015	1,996	0.035*	0.002	1,414
	(0.163)	(0.033)		(0.102)	(0.020)		(0.019)	(0.004)	
Patent families, five years before									
none	0.653***	−0.001	558	0.194	0.006	658	0.036	0.004	425
	(0.247)	(0.059)		(0.129)	(0.028)		(0.029)	(0.007)	
below the median	0.463***	−0.009	1,063	0.230**	−0.017	1,143	0.035*	0.011**	815
	(0.166)	(0.035)		(0.110)	(0.022)		(0.021)	(0.006)	
above the median	0.474***	−0.020	1,593	0.258***	−0.026	1,609	0.042**	−0.002	1,210
	(0.147)	(0.029)		(0.093)	(0.021)		(0.020)	(0.004)	

Notes: Each row is the main specification on a group of events, against the main comparison group; the groups are formed by the firm's characteristics before the onset. R&D stock over total assets: the R&D of the five years before the onset, depreciated at 15 per cent a year, over total assets at $t-1$. Intangible assets at $t-1$. Terciles of the estimation sample. Patent families: those the firm originated in the five years before, on the firms matched to patent assignees; none, or a positive stock below or above its median. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

Table G3. The response by the firm's size and financial resources

	Log R&D			Inverse hyperbolic sine of R&D			R&D over lagged assets		
	After	Before	Events	After	Before	Events	After	Before	Events
Named rivals	0.491***	−0.005	3,588	0.232***	−0.013	4,328	0.036**	0.004	3,138
(main)	(0.140)	(0.027)		(0.089)	(0.015)		(0.017)	(0.004)	
Total assets									
smallest third	0.634***	0.001	1,044	0.285**	−0.004	1,178	0.031	0.014*	883
	(0.184)	(0.038)		(0.118)	(0.022)		(0.023)	(0.007)	
middle third	0.488***	0.002	1,319	0.264***	−0.012	1,571	0.037*	0.003	1,143
	(0.145)	(0.034)		(0.100)	(0.020)		(0.019)	(0.004)	
largest third	0.299*	−0.020	1,225	0.146	−0.021	1,579	0.042**	−0.002	1,112
	(0.163)	(0.039)		(0.092)	(0.023)		(0.018)	(0.004)	
Cash over total assets									
lowest third	0.445**	0.001	876	0.243***	0.002	1,325	0.020	0.006	974
	(0.174)	(0.049)		(0.092)	(0.023)		(0.019)	(0.005)	
middle third	0.340**	0.007	1,506	0.134	−0.019	1,713	0.015	−0.002	1,207
	(0.149)	(0.036)		(0.092)	(0.022)		(0.018)	(0.004)	
highest third	0.644***	−0.021	1,206	0.313**	−0.017	1,290	0.061**	0.010	957
	(0.189)	(0.031)		(0.126)	(0.021)		(0.027)	(0.008)	
Leverage									
lowest third	0.589***	0.003	1,103	0.245**	−0.011	1,240	0.035	0.007	896
	(0.166)	(0.033)		(0.111)	(0.021)		(0.022)	(0.006)	
middle third	0.426***	−0.003	1,401	0.240**	−0.009	1,591	0.039**	0.004	1,128
	(0.162)	(0.031)		(0.102)	(0.019)		(0.019)	(0.005)	
highest third	0.394**	−0.018	1,081	0.202**	−0.017	1,494	0.035*	0.003	1,112
	(0.161)	(0.043)		(0.088)	(0.023)		(0.021)	(0.004)	

Notes: Each row is the main specification on a group of events, against the main comparison group; the groups are formed by the firm's characteristics before the onset. Total assets, cash over total assets and leverage at $t-1$, in terciles of the estimation sample. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

Table G4. The response by the firm's prior exposure to rival consolidation

	Log R&D			Inverse hyperbolic sine of R&D			R&D over lagged assets		
	After	Before	Events	After	Before	Events	After	Before	Events
Named rivals (main)	0.491***	−0.005	3,588	0.232***	−0.013	4,328	0.036**	0.004	3,138
	(0.140)	(0.027)		(0.089)	(0.015)		(0.017)	(0.004)	
Events with a full five-year history									
all of them	0.623***	0.013	1,273	0.270**	−0.002	1,634	0.043**	0.003	1,633
	(0.211)	(0.032)		(0.128)	(0.019)		(0.020)	(0.004)	
Rival consolidation in the five years before									
none	0.514**	0.023	265	0.265	0.002	358	0.032	0.003	358
	(0.261)	(0.039)		(0.167)	(0.022)		(0.028)	(0.005)	
below the median	0.766***	−0.085	408	0.421***	−0.051	526	0.056**	−0.007	525
	(0.280)	(0.062)		(0.156)	(0.037)		(0.027)	(0.006)	
above the median	0.607***	0.043	600	0.184	0.014	750	0.041*	0.008*	750
	(0.227)	(0.038)		(0.134)	(0.024)		(0.022)	(0.004)	
Years since the last rival deal									
two	0.689***	0.034	608	0.303*	0.015	785	0.055**	0.004	785
	(0.266)	(0.048)		(0.157)	(0.029)		(0.027)	(0.005)	
three or four	0.388	0.008	339	0.073	−0.019	414	0.016	0.005	413
	(0.248)	(0.042)		(0.140)	(0.026)		(0.028)	(0.006)	
five or more	0.658***	−0.003	326	0.337**	−0.008	435	0.040	0.002	435
	(0.238)	(0.039)		(0.155)	(0.023)		(0.025)	(0.005)	

Notes: Each row is the main specification on a group of events, against the main comparison group; the groups are formed by the firm's characteristics before the onset. The groups are formed on the events whose rivals' deals are observed in each of the five years before the onset. Rival consolidation: the sum, over $t-5$ to $t-2$, of the share of the firm's designated rivals that entered a consolidation in the year ($t-1$ is quiet for every event); none, or a positive stock below or above its median. Years since the last rival deal: two, three or four, or five or more (including none in the five years). After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

Appendix H. Alternative explanations

Tables H1 and H2 give the estimates of Figure 11 with their placebo periods and events, and Figure H1 plots the naming of rivals around the onset (Section VI.D).

Table H1. The alternative explanations: other deals, targets and anticipation

	Log R&D			Inverse hyperbolic sine of R&D		
	After	Before	Events	After	Before	Events
Named rivals (main)	0.491***	−0.005	3,588	0.232***	−0.013	4,328
	(0.140)	(0.027)		(0.089)	(0.015)	
Other deals at the same time						
own-deal window excluded	0.560***	0.017	2,300	0.278**	0.004	2,692
	(0.170)	(0.033)		(0.108)	(0.019)	
own deal at t not controlled	0.496***	−0.005	3,588	0.233***	−0.012	4,328
	(0.143)	(0.027)		(0.090)	(0.015)	
supplier and customer	0.369***	−0.005	3,588	0.203**	−0.010	4,328
deals controlled	(0.133)	(0.027)		(0.084)	(0.015)	
partner deals controlled	0.495***	−0.011	3,588	0.247***	−0.014	4,328
	(0.134)	(0.027)		(0.089)	(0.015)	
rivals of rivals'	0.559***	−0.024	3,588	0.185*	−0.020	4,328
deals controlled	(0.177)	(0.032)		(0.109)	(0.019)	
industry merger	0.474***	−0.004	3,588	0.214**	−0.012	4,328
wave controlled	(0.136)	(0.027)		(0.089)	(0.015)	
all of these controlled	0.366*	−0.024	3,588	0.105	−0.021	4,328
	(0.189)	(0.033)		(0.117)	(0.019)	
Becoming a target						
firms acquired from	0.473***	−0.002	3,328	0.221**	−0.011	3,948
t+1 to t+5 excluded	(0.142)	(0.028)		(0.090)	(0.016)	
Naming in anticipation						
every tie named 3	0.848***	0.017	738	0.327***	−0.006	956
or more years before	(0.246)	(0.035)		(0.120)	(0.021)	
every tie named within 2 years	0.404***	−0.005	2,345	0.204**	−0.013	2,743
	(0.125)	(0.031)		(0.083)	(0.018)	
growth of the rival	0.513	−0.033	3,588	0.258	−0.012	4,328
set controlled	(0.359)	(0.034)		(0.198)	(0.019)	

Notes: Each row changes one element of the main specification (Section VI describes each). Own-deal window excluded: events and comparison firm-years with an own consolidation from $t-2$ to t removed. Channels controlled: indicators for the channel's deals at t, $t-1$ and $t-2$. Ties: the events in which every consolidating rival had been named at least three years, or at most two years, before its deal. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

Table H2. The alternative explanations: records, definitions and comparison groups

	Log R&D			Inverse hyperbolic sine of R&D		
	After	Before	Events	After	Before	Events
Named rivals (main)	0.491***	−0.005	3,588	0.232***	−0.013	4,328
	(0.140)	(0.027)		(0.089)	(0.015)	
Years without a recorded rival						
recorded onsets only	0.652***	−0.024	2,102	0.299***	−0.019	2,643
	(0.181)	(0.031)		(0.112)	(0.018)	
recorded onsets	0.350**	0.002	2,102	0.283***	−0.006	2,643
and comparisons	(0.138)	(0.037)		(0.091)	(0.020)	
What counts as a consolidation						
deals dated by completion	0.447***	−0.001	3,645	0.172*	−0.009	4,402
	(0.160)	(0.028)		(0.096)	(0.016)	
events with a whole-firm deal	0.444***	−0.007	2,714	0.232***	−0.011	3,169
	(0.139)	(0.029)		(0.088)	(0.017)	
events with	0.483**	−0.016	723	0.247**	−0.018	837
whole-firm deals only	(0.188)	(0.043)		(0.108)	(0.029)	
The comparison						
five placebo years	0.491***	−0.025	3,588	0.232***	−0.024	4,328
	(0.140)	(0.046)		(0.089)	(0.026)	
seven quiet years	0.475***	−0.000	3,588	0.243***	−0.005	4,328
	(0.135)	(0.029)		(0.089)	(0.017)	
not-yet-exposed firms	0.491***	0.002	3,588	0.244***	0.002	4,328
	(0.135)	(0.028)		(0.089)	(0.017)	
the estimator's standard set	0.521***	−0.010	3,588	0.249***	−0.018	4,328
	(0.159)	(0.019)		(0.071)	(0.012)	

Notes: Each row changes one element of the main specification (Section VI describes each). Recorded onsets: events with a rival on record in the year before; and comparisons: comparison firm-years with a rival on record at t and $t-1$. Whole-firm deals: the events with at least one, or only, acquisitions of whole companies. After: the pooled post-period estimate, the average change in the outcome from $h=0$ to $h=5$ relative to the year before the deal, on the firm-years observed through $h=5$; Before: the pooled placebo estimate over the horizons $h=-4$ to -2. Standard errors clustered by the firm's reporting unit in parentheses. Events: the exposed firm-years at $h=0$. *, **, ***: significance at 10, 5 and 1 per cent.

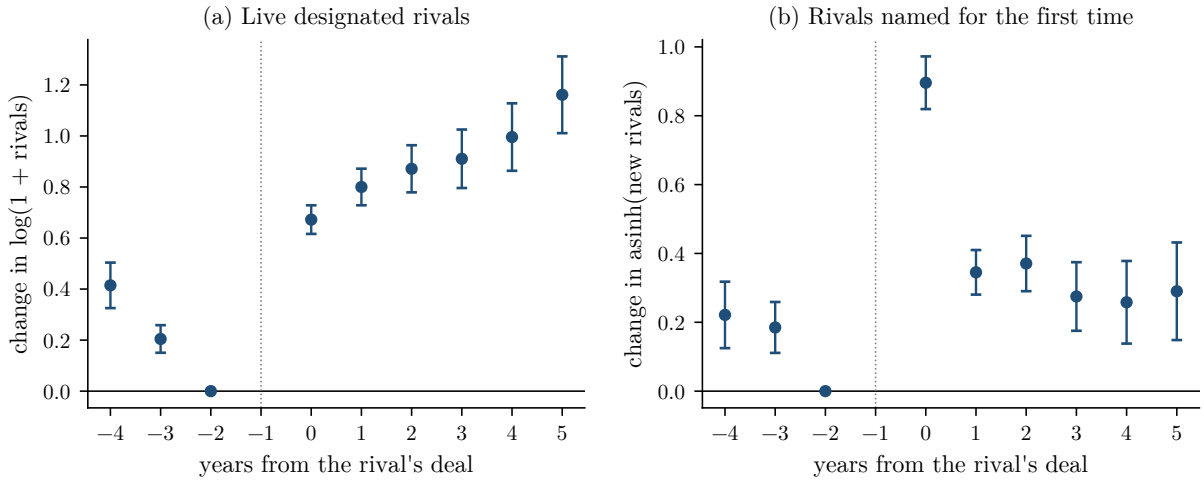


Figure H1. The naming of rivals around the onset

Notes: The main design with the firm's rival set as the outcome, on the firm-years of the log R&D sample: the log of one plus the number of live designated rivals (panel a), and the inverse hyperbolic sine of the number of rivals named for the first time in the year (panel b); the controls leave out the size of the rival set and its reporting indicators. Bars are 95 per cent confidence intervals, standard errors clustered by the firm's reporting unit; the year before the onset is the reference.

Appendix I. R&D in levels

The main estimates compare transformations of R&D. Wooldridge (2023) extends the two-way fixed-effects regression to nonlinear models of staggered treatments; for a nonnegative outcome such as R&D, he models its mean in levels with an exponential function, estimated by pooled Poisson quasi-maximum likelihood:

$$\text{(I1)}\quad E[R_{it} \mid \cdot] \;=\; \exp\Big(\alpha_{G_i} + \lambda_t + \rho\, n_{it} + \mathbf{x}_i'\boldsymbol{\kappa} + \mathbf{x}_i'\boldsymbol{\eta}_{G_i} + \mathbf{x}_i'\boldsymbol{\pi}_t + \sum_{g}\sum_{s \geq g} \tau_{gs}\, \mathbf{1}\{G_i = g,\; t = s\}\Big),$$

where R_{it} is R&D in millions of dollars, zeros included, G_i is the year of the firm's first exposure to a designated rival's consolidation (its cohort), α_{G_i} and λ_t are cohort and year effects, n_{it} indicates a year without a rival on record, $\mathbf{x}_i$ are the firm's characteristics in its first observed year, which shift the level of R&D by cohort and its trend by year, and τ_{gs} is a separate effect for each cohort in each year from its first exposure on. Because each cohort-year has its own coefficient, a firm already exposed never serves as the comparison for a later cohort. The identifying assumption is that, conditional on the characteristics, the logarithm of mean R&D would have followed parallel paths. We estimate the model with `jwdid` (Rios-Avila, Nagengast and Yotov 2022) and report its own aggregates (Table I1).

The estimator differs from the main design in three respects. Its event is the firm's first exposure, kept on whatever follows, so a firm observed only after its first exposure has no pre-exposure year and drops out: column (2) rests on 9,739 firm-years of 1,988 firms. Industry-by-year and country-by-year effects would enter as dummies, many of them for cells without any R&D, which the estimator cannot fit. And the lagged change

in R&D has no counterpart in a model of levels. The characteristics enter in the form that Wooldridge's conditional parallel-trends assumption requires; his fuller version, which also lets the effect vary with them, does not converge on this panel.

Where the two designs can be compared, they agree. With the characteristics, the average effect over every exposed firm-year is $+0.50^{**}$ log points, or 19.9^{***} million dollars. It starts near zero, +0.07 in the year of first exposure, and grows to $+0.75^{***}$ five years later, as the main estimates grow from $+0.04^{*}$ in the year of the onset to $+0.95^{***}$ in the fifth. Without the firms party to a consolidation of their own around their first exposure, the average effect is $+0.94^{***}$. Because the Poisson weighs firms by their R&D in dollars and needs no transformation of the zeros, the agreement does not come from the scale of the main estimates.

The pre-exposure years are tested jointly in two ways. The first test adds indicators for the first to fifth year before first exposure to the same estimator, whose comparison is the firms not yet and never exposed (Borusyak, Jaravel and Spiess 2024). The second, the estimator's own event-study test, gives every cohort a term in every pre-exposure year, with the year before first exposure as the reference, and so compares with the never-exposed firms only. With cohort and year effects alone, the second test rejects ($p = 0.03$) and the first rejects at ten per cent ($p = 0.07$): the firms that become exposed were already raising their R&D relative to the firms never exposed. Once the characteristics shift the trend year by year, the first test does not reject ($p = 0.11$) and the second is at the margin ($p = 0.05$); without the firms with an own deal around their first exposure, the first gives $p = 0.67$, and the second does not converge on that smaller sample.

Columns (4) to (6) drop the years without a rival on record instead of coding them as years without a rival deal, which this estimator allows because it drops firm-years one at a time rather than requiring a firm's status over a window. The average effect is similar, $+0.57^{***}$ with the characteristics, but both pre-trend tests then reject at five per cent ($p = 0.03$ and $p = 0.04$), and we do not read columns (4) and (5); without the own-deal firm-years, the first test does not reject ($p = 0.41$) and the average effect is $+0.89^{***}$. In levels, then, the response has the sign, the size and the growth of the main estimates. Its pre-exposure years pass the tests only once the firm's characteristics are controlled for, at the margin against the never-exposed firms, and with the unrecorded years coded as the main design codes them.

Table I1. R&D in levels: the extended two-way fixed effects Poisson

	Unrecorded years: no deal			Unrecorded years: excluded		
	(1)	(2)	(3)	(4)	(5)	(6)
Average effect, log points	0.454***	0.502**	0.940***	0.444***	0.565***	0.886***
	(0.148)	(0.204)	(0.183)	(0.167)	(0.194)	(0.184)
Average effect, million dollars	19.6***	19.9***	28.8***	22.3***	20.5***	28.4***
	(7.1)	(7.3)	(3.5)	(8.3)	(7.0)	(3.7)
By years since first exposure, log points						
0	0.212***	0.069	0.246***	0.268***	0.080	0.303***
	(0.074)	(0.071)	(0.066)	(0.088)	(0.072)	(0.070)
1	0.293***	0.274***	0.486***	0.336***	0.309***	0.523***
	(0.102)	(0.106)	(0.096)	(0.125)	(0.107)	(0.100)
2	0.396***	0.409***	0.605***	0.496***	0.419***	0.578***
	(0.120)	(0.135)	(0.116)	(0.152)	(0.126)	(0.124)
3	0.399***	0.427**	0.787***	0.515***	0.467***	0.722***
	(0.139)	(0.167)	(0.146)	(0.174)	(0.157)	(0.166)
4	0.476***	0.508**	0.976***	0.479**	0.592***	0.890***
	(0.160)	(0.214)	(0.199)	(0.207)	(0.214)	(0.205)
5	0.737***	0.751***	1.580***	0.665***	0.930***	1.483***
	(0.192)	(0.281)	(0.299)	(0.227)	(0.280)	(0.295)
Pre-trend tests, p-values						
Not-yet- and never-exposed firms	0.075	0.113	0.669	0.078	0.034	0.413
Never-exposed firms only	0.028	0.054	n.c.	0.009	0.043	n.c.
Observations	10,257	9,739	6,880	9,637	9,157	6,547
Firms	2,115	1,988	1,548	2,109	1,982	1,541
Indicator for unrecorded years	yes	yes	yes	–	–	–
Firm characteristics	no	yes	yes	no	yes	yes
Own-deal window excluded	no	no	yes	no	no	yes

Notes: Wooldridge's extended two-way fixed effects Poisson (Wooldridge 2023), estimated with `jwdid` (Rios-Avila, Nagengast and Yotov 2022). The treatment is the firm's first exposure to a designated rival's consolidation, and the comparison the firms not yet exposed and never exposed; R&D in millions of dollars, zeros included. (1) and (4): cohort and year effects. (2) and (5): in addition, the firm's characteristics in its first observed year (log total assets, log sales, log age, cash and capital expenditure over assets), their level and their interactions with cohort and with year. (3) and (6): columns (2) and (5) without the firms party to a consolidation in the year of first exposure or the two years before, and without the comparison firm-years with such a deal in the year or the two before. Columns (1) to (3) code a year without a rival on record as a year without a rival deal and include an indicator for such years; columns (4) to (6) drop these years. Average effect: the estimator's average over every exposed firm-year, in log points (the linear index) and in millions of dollars (the mean). Pre-trend tests: joint tests of the pre-exposure years. With the not-yet- and never-exposed firms, indicators for the first to fifth year before first exposure added to the same estimator (Borusyak, Jaravel and Spiess 2024); with the never-exposed firms only, the estimator's event-study test of the pre-exposure cohort-year terms from −5 to −2, with the year before first exposure as the reference. n.c.: the estimator does not converge. Firms: reporting units. Standard errors clustered by the firm's reporting unit in parentheses. *, **, ***: significance at 10, 5 and 1 per cent.